Storage, drought and climate change: the reliability of the canal supply from Klein Maricopoort Dam and the role of groundwater in A31E, North West Province

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Storage, drought and climate change: the reliability of the canal supply from Klein Maricopoort Dam and the role of groundwater in A31E, North West Province

Abstract

Irrigation in the A31E quaternary catchment, North West Province, appears to depend on a canal from Klein Maricopoort Dam, which carried little water for ten successive years from 1984. This desktop study assessed the reliability of the canal supply against the registered volume of 2.34 Mm³/a, the extent to which groundwater could offset its shortfalls and the response of both to climate change, and it screened flood risk and contaminant pathways. A monthly water balance of the dam (1936–2009) was run with and without a restriction rule and tested against the records of 1987–2009, and a steady-state groundwater model was run in two structures that differ in the depth of outflow. The registered volume is delivered in full in only 45% to 77% of the years, and without restriction only about 0.36 Mm³/a could have been supplied in every month of the record. Abstracting the mean shortfall beneath the irrigated land lowers the water level by about 1 to 2 m in one structure and by 5 to 21 m or more in the other. In the first structure, about 89% of this water would come from the rivers feeding the downstream dam. By about 2050, the reliability by time decreases by a further 4 to 11 percentage points. The 1:100-year flood passes the modelled reach almost unattenuated, and the water released from the dam exceeds the salinity target for irrigation in most samples. The canal supply is not reliable against the registered volume, and groundwater could offset the average shortfall in volume, but the available data cannot show whether this water would be new or whether it could sustain the scheme through a long drought.

Keywords: canal supply reliability; reservoir water balance; multi-year drought; groundwater flow model; capture; climate change; design flood; particle tracking

1. Introduction

In semi-arid regions, irrigated agriculture depends largely on water that has been stored. Irrigators in a semi-arid South African catchment have coped with past droughts mainly through the construction of water storage and the adoption of more

efficient irrigation (Lankford *et al*., 2023). The security of an irrigation supply depends on how reliably a dam can deliver water, which is described by how often the supply fails, how quickly it recovers and how severe a failure is (Hashimoto *et al*., 1982). In the Marico area of the North West Province, irrigation was the largest water use in 2000, at approximately 32 million cubic metres per year (Mm^3/a), and most of this water was drawn from surface water below dams such as Marico-Bosveld and Klein Maricopoort (DWAF, 2004:ii, 52).

In South Africa, the quaternary catchment serves as the fundamental hydrological unit for managing water resources (DWS, s.a.d). In semi-arid quaternary catchments, river flow and groundwater recharge do not respond to rainfall in simple proportion. Where about 500 mm of rain falls per year, a decrease of 10% in rainfall may reduce perennial surface drainage by half (De Wit & Stankiewicz, 2006), and groundwater in drylands is recharged episodically, mainly by heavy rainfall (Cuthbert *et al*., 2019). Dry years tend to occur in runs, and six of the eight most severe two-year droughts in southern Africa between 1901 and 1999 ended between 1983 and 1996 (Rouault & Richard, 2005). During such dry periods, groundwater sustains river flow as baseflow (Xie *et al*., 2024). Consequently, water availability in a quaternary catchment is strongly influenced by the sequence of wet and dry years and by the interaction between surface water and groundwater.

Groundwater is a crucial water resource in many semi-arid regions, because it buffers the effects of rainfall variability (Calow *et al*., 2010). In South Africa, groundwater contributes approximately 15% of the water consumed nationally (DWAF, cited by Knüppe, 2011:69), but in the Marico area it supplied 6 of the 32 Mm^3/a used for irrigation (DWAF, 2004:52). Groundwater can be abstracted sustainably only at a rate that is balanced by recharge, and over-abstraction has already lowered groundwater levels in the dolomite aquifers of the North West Province (Cobbing, 2018). Its use for irrigation is further limited by the yield of boreholes, since yields above 5 L/s are not widespread in Africa (MacDonald *et al*., 2012).

The A31E quaternary catchment is a semi-arid catchment in which irrigation depends on stored surface water. The largest part of its registered water use is not groundwater, but water that appears to be supplied by a canal from Klein Maricopoort Dam, which stands on the upstream boundary of the catchment (DWAF, 2004:50; DWS, s.a.f). A failure of this supply directly affects the water users in the catchment, and it raises the

question of whether groundwater within A31E could sustain them through such a failure.

1.1 Research problem

Klein Maricopoort Dam stores water that is generated in the catchments above A31E. Of the 3.38 Mm³/a of water use registered in A31E, 2.34 Mm³/a (69%) is registered as scheme use on 442 ha within about 15 km of the dam wall, while 0.285 Mm³/a is registered from boreholes (DWS, s.a.f). These scheme records are inferred to represent the users of the canal, because the Department lists a canal scheme for the dam and the land mapped as irrigated lies below it (DFFE, 2024; DWS, 2024).

The records show that this supply has failed for extended periods. Between October 1983 and September 1995, the dam stood below 25% of its capacity in 139 of the 144 months, and this period is referred to as the drought throughout this study. During these years the canal carried a mean of 0.25 Mm³/a, compared with a median of 2.02 Mm³/a over 72 years of record (DWA, 2012; DWS, s.a.b). The inflow to the dam did not recover when the rainfall did: it stayed below its long-term median from 1982 to 1994, although the rainfall of 1986–1995 averaged 99% of its long-term mean (Bailey & Pitman, 2016). The Department reports a yield of 1.4 Mm³/a for the dam, without a stated method and with very low confidence in the hydrological information of the Marico (DWAF, 2004:50, 83). How often, for how long and by how much the deliveries of the canal fall below the registered volume is not known.

It is equally uncertain whether groundwater within A31E could serve as an alternative source. The estimated recharge of 17.336 Mm³/a is far larger than the current groundwater use (DWS, 2021). The mapped borehole yields are low, each borehole has a single recorded water level, and no pumping tests or water-level series exist (DWS, s.a.c). The only gauged stream inside A31E was dry in 42 of the 50 months from February 1984 (DWS, s.a.b), which suggests that groundwater storage declined during the same drought that emptied the dam.

Climate change adds a further uncertainty. Warming is projected with considerably more confidence than any change in rainfall (DEA, 2013), and droughts are projected to become more frequent and longer (Trisos *et al*., 2022). For 2041–2060, the climate models assessed by the IPCC project a warming of about 1.5 to 2.9 °C for eastern southern Africa (ESAF), the region in which A31E lies, and a change in annual rainfall

of about −8.5% to +3%, with a median of about −3% (10th to 90th percentiles; calculated from Gutiérrez *et al*., 2021; Iturbide *et al*., 2020). The problem addressed in this study is that the reliability of the canal supply, the extent to which groundwater could offset its shortfalls and the sensitivity of both to climate change have not been established. Water users and the responsible authorities lack a quantified basis for drought planning.

1.2 Aim, objectives and research questions

The aim of this study is to quantify the reliability of the canal supply from Klein Maricopoort Dam, the extent to which groundwater within A31E could offset its shortfalls and the response of both to climate change, so that drought planning in the catchment can rest on the records rather than on a registered volume. The study is a desktop assessment of existing records, intended for the Department of Water and Sanitation (DWS), the North West Premier's Office and the consulting partner AGES. Flood risk along one reach of the Klein Marico River and the pathways along which contaminants could move are screened, and the transport of contaminants is not modelled.

The objectives of the study are:

1. To compile and review the existing hydrological, geohydrological and water-use information for A31E and identify its gaps.
2. To characterise the hydrology and groundwater of A31E and develop a conceptual model of how water moves through it.
3. To simulate the water balance of Klein Maricopoort Dam and its canal and quantify the reliability of supply against the registered scheme volume.
4. To assess with a groundwater flow model how far abstraction within A31E could offset shortfalls in the canal supply, and the limits that the data place on this assessment.
5. To assess the effects of climate change by about 2050 on the reliability of the canal supply and on groundwater.
6. To provide evidence-based recommendations, including priorities for data collection, to the DWS, the North West Premier's Office and AGES.

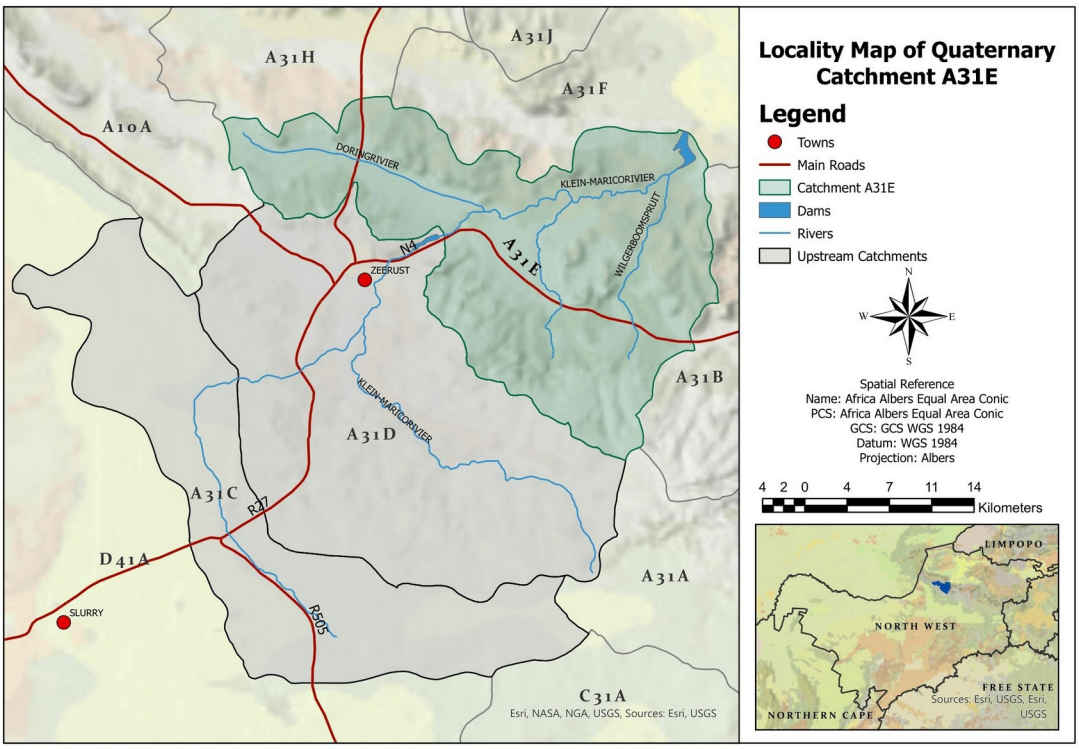
The main research question is: How reliable is the canal supply from Klein Maricopoort Dam to irrigators in quaternary catchment A31E under the historical sequence of wet and dry years, to what extent could groundwater within A31E offset the shortfalls that arise when the dam is drawn down, and how would both change under plausible climate conditions for about 2050? Five supporting questions follow from it:

1. How have the recorded inflow, the storage and the canal deliveries of Klein Maricopoort Dam behaved, and how did the water generated within A31E itself respond to the drought?
2. How reliably can the dam supply the registered scheme volume when its water balance is simulated over the historical record, and how sensitive is this reliability to the restriction of deliveries and to the demand?
3. What can a groundwater flow model built on the available records show about abstraction intended to offset a shortfall of the canal, and where would the pumped water come from?
4. How would the warmer and probably drier conditions projected for about 2050 alter the reliability of the canal supply and the availability of groundwater?
5. How large are the floods along the irrigated reach, and along which paths could contaminants from the land reach the receptors?

2. Study Area

2.1 Location and the two dams

Quaternary catchment A31E lies in the North West Province, east of Zeerust, and covers approximately 602 km² (Bailey & Pitman, 2016). It drains to the Marico River, part of the upper Limpopo River basin (Jury, 2016), within the Limpopo-Olifants Water Management Area (DWS, 2023). The Klein Marico River enters A31E at the wall of Klein Maricopoort Dam and leaves the catchment through Kromellenboog Dam (Map 1). The reservoir of the upper dam and its whole drainage area lie outside the catchment, and none of the runoff generated within A31E can reach Klein Maricopoort Dam. The water stored in the upper dam comes entirely from outside A31E, while the runoff of A31E flows to Kromellenboog Dam, which is operated for an irrigation scheme in the downstream catchment A31F (DWAF, 2004:50, 57).



Map 1: Locality of quaternary catchment A31E, showing the Klein Marico River, Klein Maricopoort Dam at the upstream boundary, Kromellenboog Dam at the outlet, the upstream catchments A31D and A31C, Zeerust and the main roads. Inset: position in the North West Province.

The canal leaves the upper dam at its wall, and the Department lists an irrigation scheme for the dam that is supplied “from the canal system” (DWS, 2024). The dam also releases water to the river. It was raised in 1966, and its survey of 1982 gives a capacity of 7.07 Mm³ at full supply (DWA, 2012).

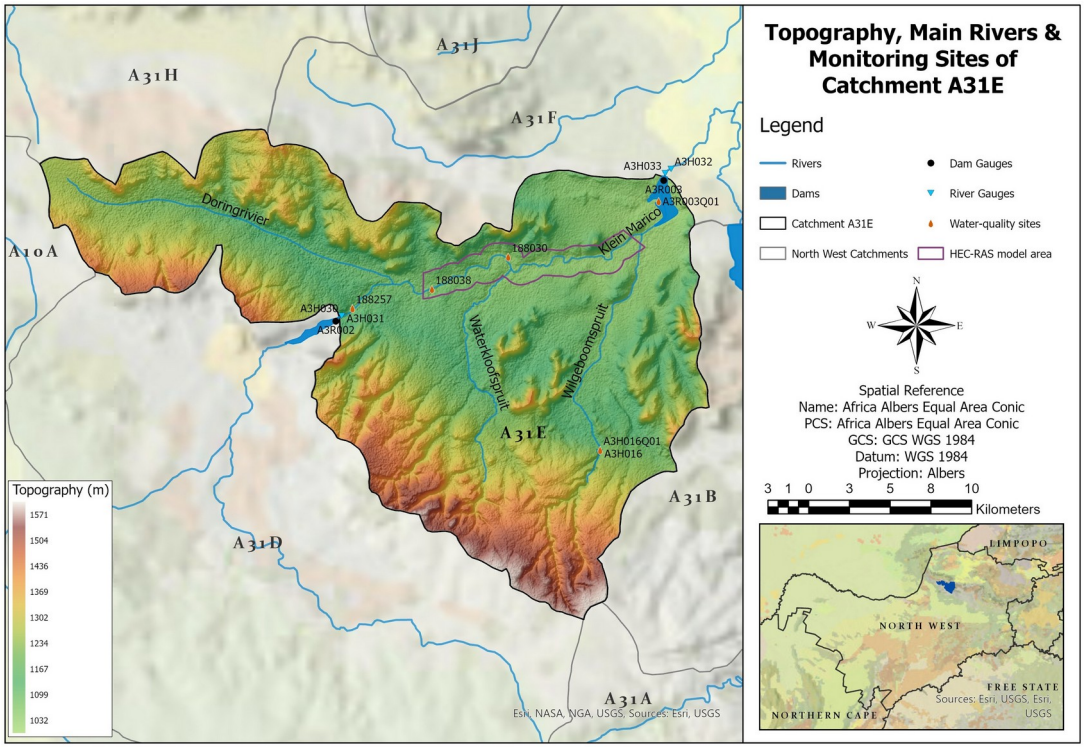
2.2 Climate

The climate of the Marico area is semi-arid, with rainfall that is “strongly seasonal with rainfall occurring as thunderstorms in summer” (DWAF, 2004:18). The mean annual precipitation (MAP) of A31E is 597 mm in the Water Resources of South Africa, 2012 study (WR2012), which takes its rainfall from the series of rainfall zone A3B (Bailey & Pitman, 2016:14). The mean annual evaporation of the evaporation zone of A31E is 1 901 mm (Bailey & Pitman, 2016).

2.3 Topography and drainage

A31E falls from the high ground in the south to Kromellenboog Dam, and the valley of the Klein Marico River crosses the northern part of the catchment (Map 2). Four streams drain A31E. The Klein Marico River runs the length of the valley, the Doringrivier drains the western arm and joins it below the dam wall, the

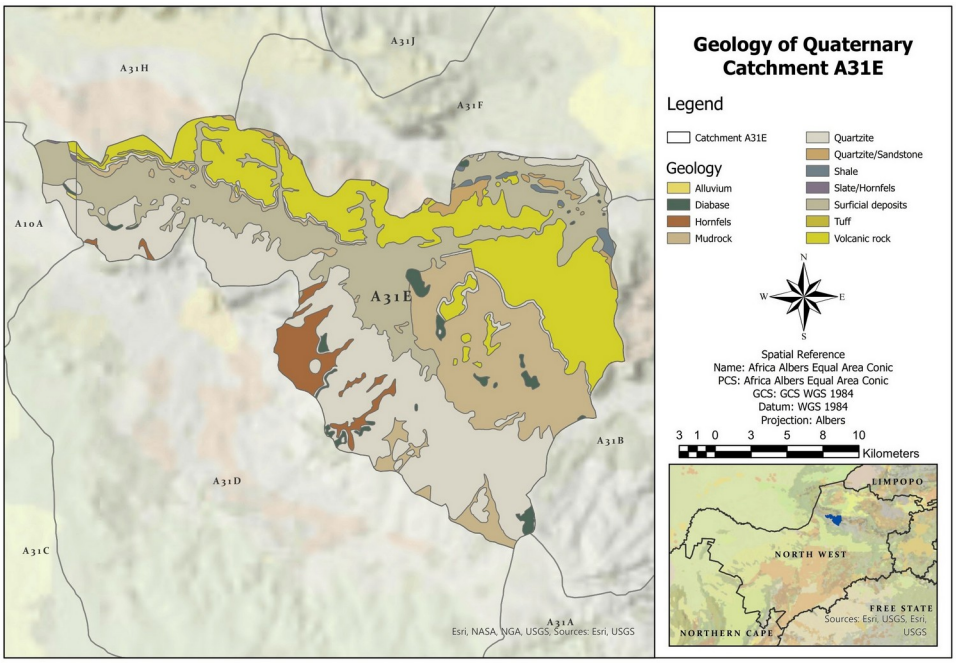
Waterkloofspruit joins it further downstream, and the Wilgeboomspruit reaches the river at the head of Kromellenboog Dam. The modelled naturalised runoff of A31E is 7.21 Mm³/a (Bailey & Pitman, 2016).



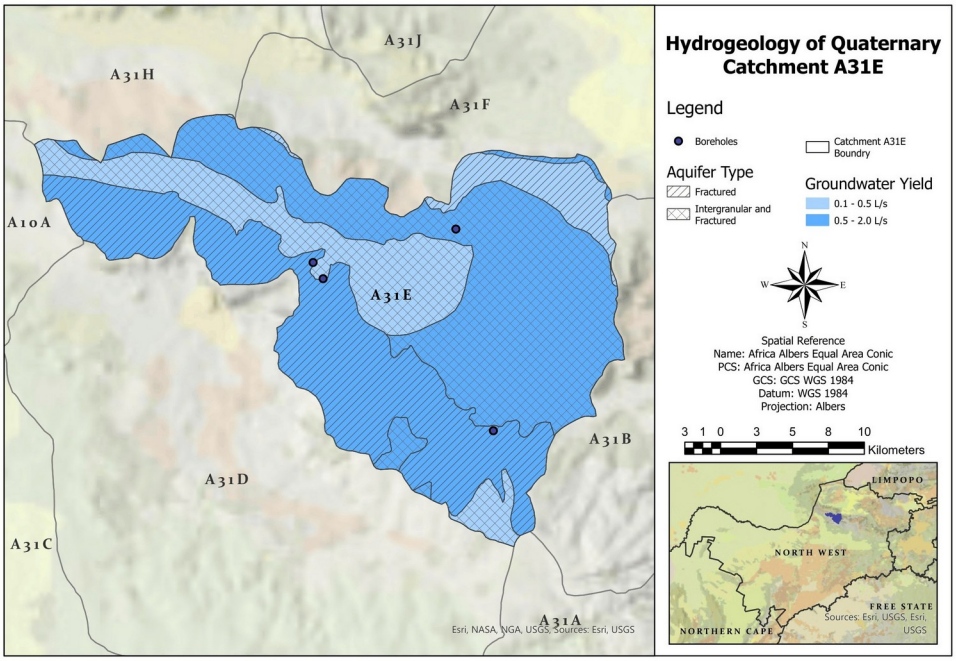
Map 2: Topography, main rivers and monitoring sites of A31E, with the area of the HEC-RAS model along the Klein Marico River. The stations at the wall of Klein Maricopoort Dam lie within 400 m of one another, and only some are labelled.

2.4 Geology and hydrogeology

A31E is underlain by rocks of the Pretoria Group, which are cut by diabase and partly covered by surface deposits (Map 3; Michaluk & Keyser, 1991; Walraven, 1981). Quartzite and sandstone, mainly of the Timeball Hill Formation, form the high ground. The andesitic lava of the Hekpoort Formation, diabase, mudrock, shale and hornfels make up the remaining bedrock, and surface deposits lie mainly along the valleys. Groundwater occurrence depends primarily on geology, weathering and effective rainfall (MacDonald *et al*., 2012:2). On the hydrogeological map, A31E is classified as an intergranular and fractured aquifer and a fractured aquifer, and karst aquifers do not occur (Map 4; Barnard, 1999). The National Groundwater Archive (NGA) holds the borehole records of the catchment (DWS, s.a.c). The Department judged that groundwater in the Marico can meet domestic supplies “but not bulk water requirements for purposes such as irrigated agriculture” (DWAF, 2004:55–56).



Map 3: Geology of A31E, showing the rock types of the 1:250 000 geological maps (Michaluk & Keyser, 1991; Walraven, 1981). The rock types are combined into four groups: quartzite and sandstone; volcanic rocks, diabase and tuff; mudrock, shale and hornfels; and surface deposits, which include the alluvium.



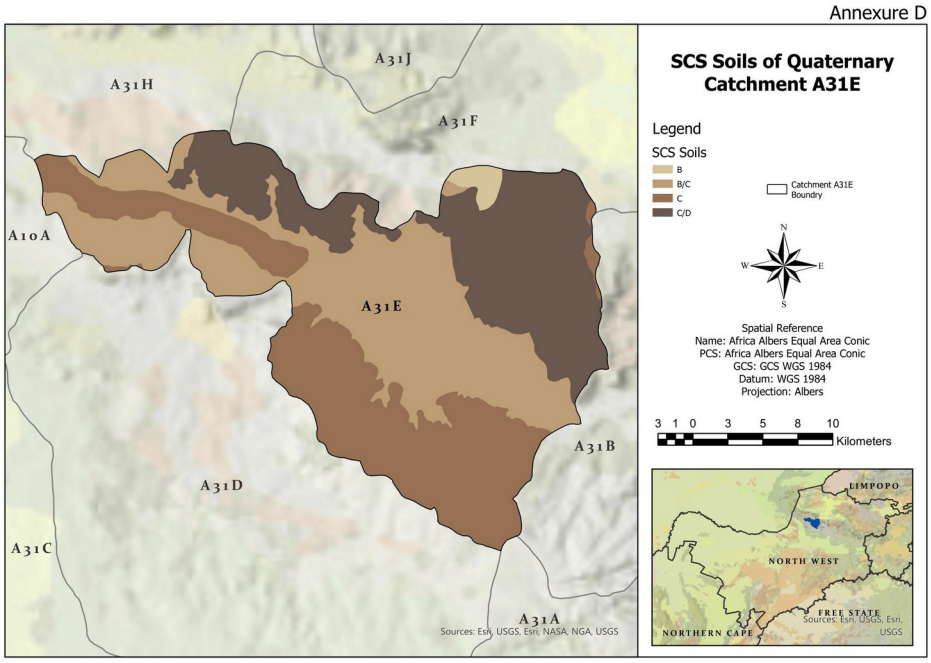
Map 4: Aquifer types and mapped borehole-yield classes in A31E (Barnard, 1999), with the four boreholes that have a static water level (DWS, s.a.c). The yield classes are mapped medians, not measured yields.

The Groundwater Resource Assessment II (GRAII) gives the following modelled regional values for A31E (DWS, s.a.a). Transmissivity, which is how easily the full thickness of an aquifer passes water, is 22.81 m²/d. Storativity, which is the water released per m² of aquifer when its level falls by 1 m, is 0.00998 in the weathered zone

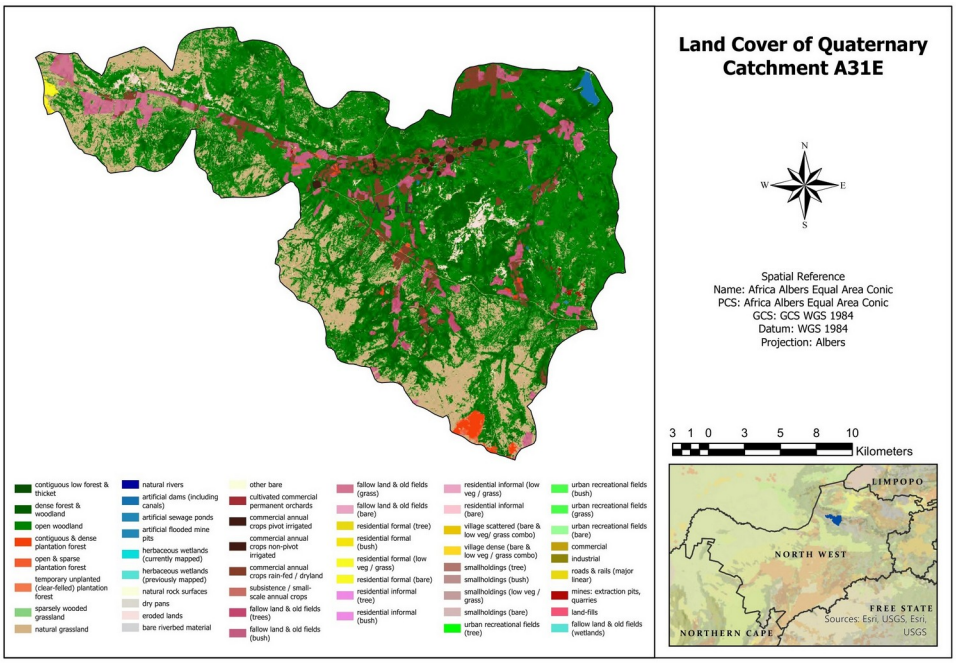
and 0.00014 in the fractured zone below it. The weathered zone is 25.36 m thick and the fractured zone 182.14 m, a total of 207.5 m. Recharge, which is the part of rainfall that reaches the groundwater, is 4.84% of rainfall, about 29 mm/a.

2.5 Soils and land cover

The soils of A31E are assigned to the hydrological soil groups B, B/C, C and C/D, which rank soils according to their runoff potential when thoroughly wet, from moderately low in group B to high in group D (Map 5; Schulze & Schütte, 2023; USDA-NRCS, 2009:7-2). The land cover of 2022 is classified from satellite images and is not surveyed on the ground (Map 6; DFFE, 2024). It consists of natural woodland and grassland, cultivated land and a small area mapped as irrigated land in the strip below the dam.



Map 5: Hydrological soil groups of A31E according to the SCS classification as applied in South Africa (Schulze & Schütte, 2023).



Map 6: Land cover of A31E in 2022 (DFFE, 2024).

2.6 Water use and water quality

The national register of water use records entitlements and not the actual abstraction (DWS, s.a.f). Of the 3.38 Mm³/a registered in A31E, 69.3% is registered as “scheme” use, 22.2% as river abstraction and 8.4% as borehole use, and 98.9% of the total is for irrigation.

The Department has described return flows from Zeerust and from irrigation that are “being discharged into the Klein Marico River” (DWAF, 2004:86), and it noted that their nutrient and salt content limits their re-use (DWAF, 2004:57). Since Zeerust and the irrigation of A31D lie above the dam, these return flows are inferred to enter the river upstream of it. Water quality was monitored at seven sites in A31E between 1966 and 2023 (DWS, s.a.e). These records are described in Section 5.1.4 and screened in Section 6.8.

3. Literature foundation

3.1 Reliability of supply from a small dam

The performance of a water-supply system is judged by more than the average volume that it delivers. Hashimoto *et al*. (1982:14–17) proposed three criteria for this purpose. Reliability describes how often the system is in a satisfactory state, resiliency describes how quickly it recovers once it has failed, and vulnerability describes how

severe a failure is. All three criteria depend on a threshold that separates a satisfactory delivery from an unsatisfactory one. In this study, a failure is a period in which the canal delivers less than the registered scheme volume, so that the criteria answer directly how often, for how long and by how much deliveries fall short. Reliability of the supply is not the same as the resilience of the irrigators, who may plant a smaller area or apply less water when the supply decreases (Lankford *et al*., 2023).

A small dam can carry a scheme through one dry year, but not through a run of dry years. The relationship between rainfall and runoff can shift during a multi-year drought (Saft *et al*., 2015), so that a relationship calibrated on wetter years may overestimate the inflow in precisely the years that determine reliability. Evaporation adds to this problem, since the loss increases with the surface area of the water and is expected to increase further as the climate warms, and it is often overlooked in the water accounting of small reservoirs (Aminzadeh *et al*., 2024). The registered volume is used as the threshold, because it is the volume that the scheme is entitled to receive, but it differs from both the water gauged at the head of the canal and the water used on the fields (Section 6.1).

The only planning assessment found for the dam is the Department's strategy report for the Marico, which gives a yield without a stated method, does not include the dam in its proposed systems model and does not mention the drought at the dam (DWAF, 2004:50, 83–84). Its finding that the Marico sub-area was "in balance" while the Reserve was not implemented (DWAF, 2004:53, 93) concerns the whole sub-area, and it says nothing about how often the supply of this particular dam fails.

3.2 Groundwater as a fallback, and modelling with few data

Groundwater security during drought depends on the water stored in an aquifer, on access to this water through boreholes and on the demand placed on it (Calow *et al*., 2010:246). A source that is reliable for domestic supply may fail under the larger demand of irrigation. Sustainable groundwater use should be based on capture, which is the decrease in discharge and increase in recharge that pumping brings about, rather than on recharge alone (Seward *et al*., 2006:473). Where groundwater discharges to rivers, capture is largely a loss of streamflow, and it supplied about 0.85 of long-term withdrawals in the aquifer systems reviewed by Konikow and Leake (2014). This loss arrives later than the pumping and continues after it stops (Leake &

Barlow, 2013), so that a balance of recharge and use cannot show which discharge an abstraction would reduce. In A31E, access to groundwater appears to be the main limitation, because the irrigated land lies on surface deposits of unknown thickness in the lower mapped yield class (Section 6.1).

Where records are sparse, models are often calibrated over short periods. Ebrahim *et al*. (2019) calibrated a groundwater model of the hard-rock Hout catchment over five years and estimated the unmetered pumping from satellite data. A model calibrated over a few years may not represent a drought that is longer than its calibration period. For Klein Maricopoort Dam, the monthly reservoir record covers 1936 to 2010 and includes the long drought, and the simplest defensible approach uses this record directly. For groundwater the records are too few to identify a single conceptual model, and several should then be carried forward (Barnett *et al*., 2012:26), because calibrated models that differ little in their heads can differ greatly in their predictions (Troldborg *et al*., 2007:843).

3.3 Climate change and the supply

The projections used in this study are those assessed by the IPCC in its Sixth Assessment Report. The Atlas of this report gives regional results for the global models of the sixth Coupled Model Intercomparison Project (CMIP6) and for the regional models of CORDEX-Africa (Gutiérrez *et al*., 2021). A31E lies in the reference region of ESAF (Iturbide *et al*., 2020), about 1° east of its boundary with western southern Africa. The projections for ESAF are used because the catchment lies inside this region, but its regional mean is weighted towards the wetter east. The clearest rainfall signal over ESAF is a later start of the summer rains, which Howard and Washington (2020) linked to a later seasonal breakdown of the Congo Air Boundary. For 2041–2060, the CMIP6 models under an intermediate emission scenario (SSP2-4.5) give a median change of about −10% in the rainfall of September to November, and 91% of the models are drier (calculated from Gutiérrez *et al*., 2021; Iturbide *et al*., 2020, 2022). The IPCC also projects an increase in agricultural and ecological drought in southern Africa (Ranasinghe *et al*., 2021:1771).

Projections that were downscaled for South Africa are less uniform. The Green Book projects that the warming may exceed 3 °C over the northern parts of the country by 2021–2050 under high emissions, and it projects more rain over the central interior but

drier conditions in the north-east in most of its runs (Engelbrecht, 2019:12, 37). The ensembles agree on strong warming and differ on rainfall. Because perennial drainage responds more than proportionally to rainfall where about 500 mm falls per year (De Wit & Stankiewicz, 2006), even a modest drying could reduce the inflow to the dam by a larger fraction. The wettest tenth of the CMIP6 models gives about +0.5% to +3% more annual rain (calculated from Gutiérrez *et al*., 2021). The climate cases of this study are drawn from the median and the dry end of the projections.

3.4 Design floods and the pathways of contaminants

Design floods in South Africa are estimated by three groups of methods (Smithers, 2012:634). Statistical methods analyse observed floods, deterministic methods convert a design rainfall into a flood, and empirical methods are based on the largest floods recorded in a region. Statistical analysis requires a record of observed floods that is “adequate in both length and quality” (Smithers, 2012:635), which the monthly records of A31E do not contain. The Standard Design Flood (SDF) is a deterministic method whose runoff coefficients were calibrated on 29 catchments (Alexander, cited by Smithers, 2012:639; SANRAL, 2013:3-43). It exceeded statistical estimates by 60% on average and is regarded as conservative (Smithers, 2012:639). Its results are therefore checked against the empirical formula of Midgley and Pitman (cited by SANRAL, 2013:3-61–3-62) and against the Regional Maximum Flood (RMF), which is an envelope of the largest recorded peaks and carries no return period (Kovács, 1988; SANRAL, 2013:3-65).

Contaminants are approached through the source–pathway–receptor model, in which water is at risk only where a source releases a contaminant, a pathway carries it and a receptor receives it (Mbae *et al*., 2024). The pathway through the ground can be traced by particle tracking. The program MODPATH follows particles through the flows calculated by MODFLOW and derives their times of travel from the porosity of the material (Pollock, 2016:2–3). Particle tracking shows where water goes and roughly how long it takes, but not how a dissolved substance spreads or reacts.

The literature provides the criteria, the reasons why multi-year droughts and evaporation govern a small dam, a projected range of climate change and methods for screening floods and pathways. It does not provide a measure of how often, for how long and by how much the deliveries from Klein Maricopoort Dam fall short, how this

would change by about 2050, or whether groundwater in A31E could make up part of a shortfall. This study addresses that gap.

4. Conceptual model

4.1 The system and its two waters

The system studied is the land between two dams, together with the upper dam that supplies it (Figure 1). Water enters this system in two ways, which are referred to as the two waters. River flow from the catchments upstream is caught in Klein Maricopoort Dam, and rain falls on A31E itself. The first water feeds the canal and the irrigated land, while the second can reach neither the upper dam nor the canal. Path 1 carries the water from outside A31E through the upper dam, where evaporation removes water but not the salts dissolved in it, to the canal and the river. Path 2 begins with the rain on A31E and runs over the ground to the streams or down to the groundwater. The two paths meet below the canal, on the irrigated land, where canal water drains to the river and the groundwater, and in Kromellenboog Dam, which receives both. The inflow to the upper dam is taken as recorded and serves as the condition at the upstream boundary.

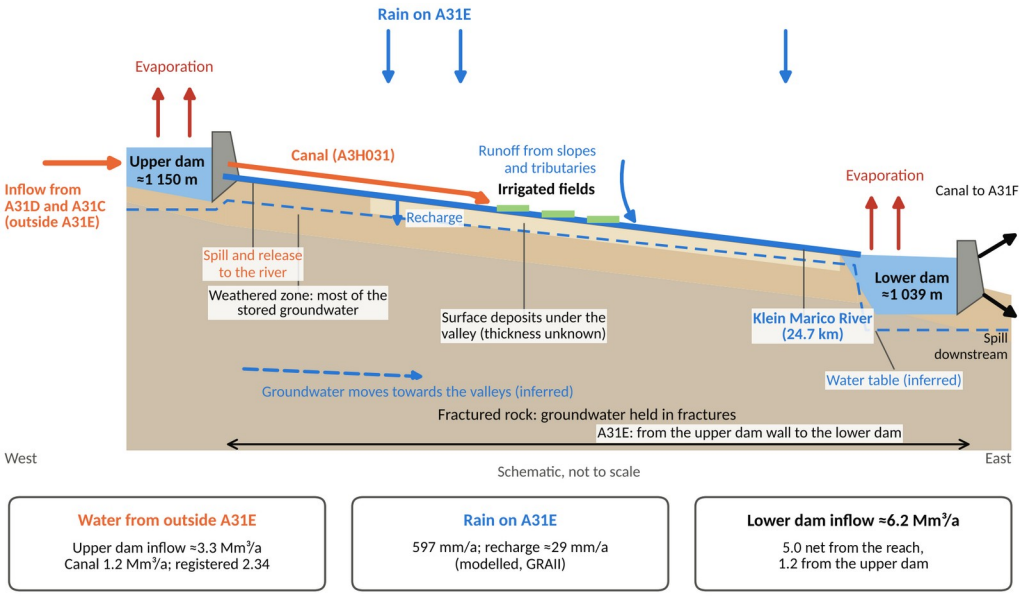


Figure 1: Schematic section of A31E along the Klein Marico River, from the wall of Klein Maricopoort Dam to Kromellenboog Dam, showing the water that enters from outside A31E (orange) and the water from rain on A31E itself (blue). Quantities are means for hydrological years 1987–2009; not to scale.

4.2 The upper dam and the canal

Water leaves the upper dam by four routes, namely evaporation, the canal, the spill and the release to the river, and no supply to Zeerust appears in the record. Beyond the gauge at its head, the canal is not measured. Its losses, the fields that it serves and the water that returns to the river or sinks to the groundwater are unknown, but none of this water can return to the upper dam. In a year of median inflow, a delivery close to the registered volume lowers the storage, while in a wet year most of the water spills, because the dam is already full. The storage of the dam and the inflow from upstream respond over several years, and these slow parts of the system decide whether the canal can deliver.

4.3 Rain on A31E, groundwater and the lower dam

The runoff of A31E is gathered by the tributaries and ends in Kromellenboog Dam. Rain that does not run off recharges the groundwater, which is stored mainly in the weathered zone. That this groundwater moves from the high ground towards the valleys is an inference from four single water levels (DWS, s.a.c). Groundwater leaves the catchment through boreholes, as discharge to the streams, by uptake of vegetation and possibly as underground outflow (DWS, 2021).

Kromellenboog Dam receives the water of both paths. The net contribution of the reach between the dams is mostly runoff and groundwater discharge of A31E, but it also contains any canal water that returns to the river from the fields, less the water abstracted from the river.

4.4 Sources, pathways and receptors of contaminants, and floods

The same conceptual model shows the routes along which dissolved substances would travel (Section 3.4). Two main sources can be identified. The first is the return flow from Zeerust and from irrigation and mines above the dam, which is inferred to enter the river upstream of it (Section 2.6) and whose nutrient and salt content limits its re-use (DWAF, 2004:57). The second is the irrigated land below the dam, where water not used by crops carries salts, and anything applied to the fields, back to the river or down to the groundwater. The rest of the catchment had a population of only 936 in 2011 and contains no large urban source (DWS, 2021:156).

These sources reach the receptors along path 1 and path 2. Below the fields, groundwater can discharge to the river only where the water table stands above the river bed, and where it stands deeper, the water leaves by evaporation and uptake by plants instead. Whether the water table stands above the river bed separates the two model structures of Section 5.2.2.

Four receptors follow from these paths: the users of the canal, who receive path 1 water; the groundwater under the irrigated land, which this study assesses as a backup to the canal and for which no quality record exists; the river between the dams; and Kromellenboog Dam, which receives both paths. The river between the dams is an ecological flow site whose category, C (moderately modified), is to be maintained (DWS, 2021:130; Kleynhans & Louw, 2007:A2-11). The DRASTIC index indicates how easily water from the land surface reaches the groundwater (Aller *et al*., 1987:13). The index is a relative evaluation tool that does not replace on-site investigation (Aller *et al*., 1987:11, 20), and it says nothing about the surface deposits in particular.

Floods follow the same two paths. A flood from upstream enters as spill over the wall of the upper dam, the tributaries add floods from A31E itself, and Kromellenboog Dam receives both. The Department identified the spillway of Klein Maricopoort Dam as “a major risk” and the spillway capacity of Kromellenboog Dam as “a major risk factor” (DWAF, 2004:170–171).

5. Methodology

The methodology is described in four steps, namely the data, their preparation, their processing within software and their analysis (Sections 5.1 to 5.4). Each step deals in turn with the water balance of the dam, groundwater, floods, and water quality and climate. All the methods rest on the conceptual model of Section 4, and the water balance uses monthly records.

5.1 Data

5.1.1 Dam water balance

The station records of the Department give one value per month with a quality code (DWS, s.a.b). The reservoir record gives the storage, outflows, evaporation and rain of Klein Maricopoort Dam for every month from 1936 to 2010 (DWA, 2012), and the

WR2012 study supplies the rainfall of zone A3B and the patched inflow (Bailey & Pitman, 2016). Evaporation at the dam is measured with a Symons pan (S-pan). The inflow is not gauged but calculated from the dam's own water balance (Bailey & Pitman, 2016:30; DWA, 2012). The only record of runoff generated inside A31E is that of the Wilgeboomspruit (station A3H016), which drains 39 km² and ends in 1992 (DWS, s.a.b). The monthly range of each record is listed in Table 1.

Table 1: Stations and records used in this study, with the range of their monthly data and, in brackets, the number of months (Bailey & Pitman, 2016; DWA, 2012; DWS, s.a.b).

Station	What it measures	Monthly data
A3R002	Spill over the wall of Klein Maricopoort Dam	1936/06–2026/06 (1 081)
A3R002	Storage in the dam (reservoir record)	1936/06–2010/11 (894)
A3H031	Volume in the canal	1936/06–2026/05 (1 080)
A3H030	River below the dam (low flows only)	1987/02–2011/05 (292)
A3H042	River below the dam	2012/01–2026/06 (174)
A3E003	Rainfall at the dam	1967/09–2025/02 (690)
A3E003	S-pan evaporation at the dam	1967/10–2025/02 (689)
A3H016	Flow in the Wilgeboomspruit (39 km²)	1965/12–1992/08 (321)
A3R003	Spill over the wall of Kromellenboog Dam	1955/02–2026/07 (858)
A3H033	Volume in the Kromellenboog canal	1955/02–2026/07 (858)
A3H032	River below Kromellenboog Dam	1979/12–2026/07 (560)
Zone A3B	WR2012 rainfall (not a station)	1920/10–2010/09 (1 080)

The Department assigns a quality code to every month. Table 2 gives the share of months in each code for the gauges at Klein Maricopoort Dam, and Table 3 gives it for the Wilgeboomspruit and the gauges at and below Kromellenboog Dam. The spill and canal records at the upper dam have about 74% good continuous data. The S-pan record has 37.0% good continuous data and 43.5% estimates, and the river gauge A3H042 has 5.2% good continuous data.

Table 2: Data quality of the monthly records at Klein Maricopoort Dam: share of the months of each record in each DWS quality code, in per cent. The number of months of each record is given in brackets (calculated from DWS, s.a.b).

Quality code	A3R002 spill (1 081)	A3H031 canal (1 080)	A3H030 river (292)	A3H042 river (174)	A3E003 rainfall (690)	A3E003 S-pan (689)
Good continuous data	73.7	73.8	71.9	5.2	61.6	37.0
Good edited data	6.0	1.1	0.0	21.3	0.0	0.0
Good monthly reading	0.0	0.0	0.0	0.0	21.7	0.2
Audited gauge plate readings	4.4	0.7	0.0	0.0	0.0	0.0
Height derived from flow	0.0	9.6	0.0	0.0	0.0	0.0
Gap filled data	2.4	0.0	0.0	0.0	0.0	0.0
Audited estimate	0.4	0.0	2.1	0.0	0.0	40.1
Minimum value	0.0	0.0	5.1	0.0	0.0	0.0
Above rating	0.0	0.0	7.2	0.0	0.0	0.0

Quality code	A3R002 spill (1 081)	A3H031 canal (1 080)	A3H030 river (292)	A3H042 river (174)	A3E003 rainfall (690)	A3E003 S-pan (689)
Permanent gap	2.9	3.5	2.7	10.3	0.0	4.4
Data missing	1.2	1.3	0.0	36.2	8.6	3.8
No info for stage/discharge determination	0.1	0.0	0.0	0.0	0.0	0.0
Data unreliable	0.0	0.2	0.0	0.0	0.0	0.6
Negative	0.0	0.0	0.0	0.0	0.0	0.2
Undefined code 8	0.3	0.0	0.0	0.0	0.0	0.0
Previous week's level used as estimate	0.5	0.0	0.0	0.0	0.0	0.0
Edited and checked – still unaudited	4.8	2.0	0.0	17.2	0.0	0.0
Checked – still unaudited	0.0	4.8	0.0	0.0	7.0	4.9
Checked estimate – still unaudited	0.7	0.6	0.0	0.0	0.0	3.5
Unaudited	0.2	0.0	10.3	0.0	1.2	5.5
Good edited unaudited	2.5	2.4	0.7	9.8	0.0	0.0

Table 3: Data quality of the monthly records of the Wilgeboomspruit and at and below Kromellenboog Dam: share of the months of each record in each DWS quality code, in per cent. The number of months of each record is given in brackets (calculated from DWS, s.a.b).

Quality code	A3H016 Wilgeboomspruit (321)	A3R003 spill, Kromellenboog (858)	A3H033 canal, Kromellenboog (858)	A3H032 river below Kromellenboog (560)
Good continuous data	79.4	66.2	69.5	51.3
Good edited data	0.0	8.6	9.3	15.7
Audited gauge plate readings	0.0	7.8	1.1	0.7
Audited estimate	0.0	0.0	1.4	1.4
Minimum value	1.9	0.0	0.0	0.0
Above rating	0.0	0.0	0.1	0.4
Permanent gap	18.4	1.4	6.5	11.3
Data missing	0.3	0.2	1.3	2.0
No info for stage/discharge determination	0.0	0.4	0.0	0.0
Data unreliable	0.0	0.1	0.0	0.0
Unknown	0.0	0.0	0.1	0.0
Undefined code 8	0.0	4.7	0.0	0.0
Previous week's level used as estimate	0.0	2.0	0.0	0.0
Edited and checked – still unaudited	0.0	5.8	6.9	12.0
Checked – still unaudited	0.0	0.0	0.7	0.4
Checked estimate – still unaudited	0.0	0.1	0.1	0.2
Good edited unaudited	0.0	2.7	3.0	4.8

5.1.2 Groundwater

The borehole records are those of the NGA (DWS, s.a.c). The NGA holds 425 boreholes inside A31E. Each of the 266 boreholes with a water level has a single level, measured between 1910 and 1994, and four of these are recorded as static water levels, three of them during the drought (Table 4). No water-level series or pumping tests exist.

Table 4: The four boreholes inside A31E with a static water level, with the date of the level (DWS, s.a.c), and the mapped classes in which they lie (measured in GIS).

Borehole	Date of level	Depth to water (m)	Rock group	Aquifer type	Mapped yield class	Soil group
2526AD00180	11 May 1994	30.00	Volcanic rocks, diabase and tuff	Intergranular and fractured	0.5–2.0 L/s	B/C
2526CA00602	9 July 1990	9.24	Quartzite and sandstone	Fractured	0.5–2.0 L/s	B/C
2526CA00864	10 July 1990	20.23	Surface deposits	Intergranular and fractured	0.1–0.5 L/s	B/C
2526CB00135	25 November 1974	10.97	Quartzite and sandstone	Fractured	0.5–2.0 L/s	C

The mapped layers are the 1:250 000 geological maps (Michaluk & Keyser, 1991; Walraven, 1981), the 1:500 000 hydrogeological map with its borehole-yield classes (Barnard, 1999), the hydrological soil groups (Schulze & Schütte, 2023) and the national land cover of 2022 (DFFE, 2024). The regional transmissivity, storativity, thickness and recharge are those of the GRAII (DWS, s.a.a; Section 2.4), and the Reserve determination gives the recharge, use and baseflow of A31E (DWS, 2021).

5.1.3 Flood

The terrain of the flood model is the 30 m Shuttle Radar Topography Mission (SRTM) elevation model of February 2000 (Farr *et al*., 2007), and the roughness follows the land-cover groups of the national land cover of 2022 (DFFE, 2024). The design rainfall, the runoff coefficients and the flood formulas are those of the drainage manual of the South African National Roads Agency (SANRAL, 2013), and the regions of the Regional Maximum Flood are those of Kovács (1988).

5.1.4 Water quality and climate

The water-quality records are the analyses of the Department at seven sites in A31E (DWS, s.a.e). The sites, in order along the Klein Marico River, with the period of each record and the number of samples with an electrical conductivity (EC), are: A3H030,

the river at the weir below Klein Maricopoort Dam (1989–1993; 56 samples); A3H042Q01, the river below the dam (2012–2023; 81); 188257, the river at Kalkdam, Enzelsberg road (2005–2007; 3); 188038, the river at Nooitgedacht (2004–2005; 3); 188030, the river at Dammenbrug (2004; 1); A3R003Q01, Kromellenboog Dam near the wall (1966–2023; 248); and A3H016Q01, the Wilgeboomspruit (1971–1991; 54). Path 1 is sampled below Klein Maricopoort Dam and at the three sites along the reach, path 2 in the Wilgeboomspruit, and both paths in Kromellenboog Dam. EC, which measures the dissolved salts, sulphate, chloride, sodium, the sodium adsorption ratio (SAR), inorganic nitrogen and phosphate are screened. The guideline values are the upper limits of the target range for irrigation and the boundaries of the trophic classes for aquatic ecosystems (DWAF, 1996a; DWAF, 1996b).

The climate cases use the CMIP6 projections of the IPCC Atlas for ESAF for 2041–2060 under SSP2-4.5 (Gutiérrez *et al*., 2021; Iturbide *et al*., 2020, 2022).

5.2 Data preparation

5.2.1 Dam water balance

Rainfall and flow are counted by hydrological year, which runs from October to September and is named after the calendar year in which it begins. Months that were partly recorded are treated as gaps, and a yearly total is counted only for a year in which every month has a value. Three periods are doubtful although their codes are good. These are the calculated inflow of 1975 to 1980, which totals 296 Mm^3 against 186 Mm^3 at Kromellenboog Dam downstream, the spill record of 1977 to 1979, and the canal record of 2018 to 2021, which repeats one flow rate (DWA, 2012; DWS, s.a.b).

The water-surface area follows from the relation between the gauge reading and the contents of the reservoir since the survey of 1982, which gives the capacity of 7.07 Mm^3 at full supply on which the reservoir record is based. Evaporation is calculated from the twelve mean monthly S-pan depths, because 43.5% of the pan months are estimates, with pan factors of 0.82 to 0.89 that were back-calculated from the Department’s evaporation volumes. The release to the river is taken as the mean gauged release in each band of storage (Figure 2, upper panel).

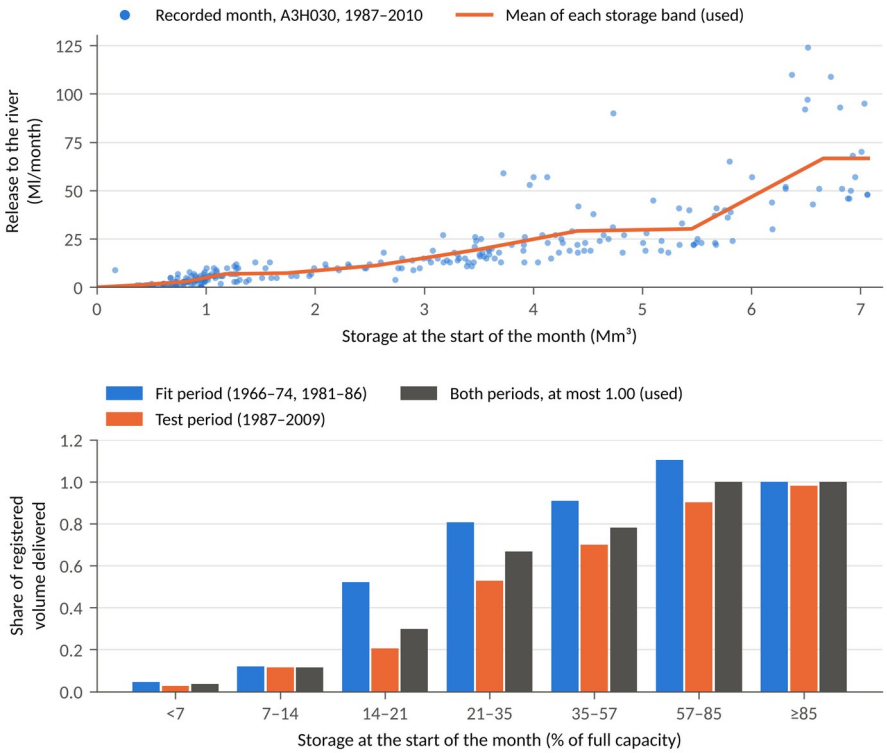


Figure 2: Upper panel: monthly release to the river below Klein Maricopoort Dam (DWS station A3H030) against the storage at the start of the month, 1987–2010 (259 months without spill), with the mean of each storage band. Lower panel: share of the registered volume that the canal received in each storage zone, in the fit period (hydrological years 1966–1974 and 1981–1986), in the test period (1987–2009) and in both together (the rule used).

The canal is asked to deliver the registered volume of 2.34 Mm³/a, which it carried when the dam was full (DWA, 2012; DWS, s.a.b). This volume is divided over the year according to the pattern of Figure 3, which is taken from the canal record A3H031 for the hydrological years 1997–2014 without 2010. About a third of the volume falls from June to September.

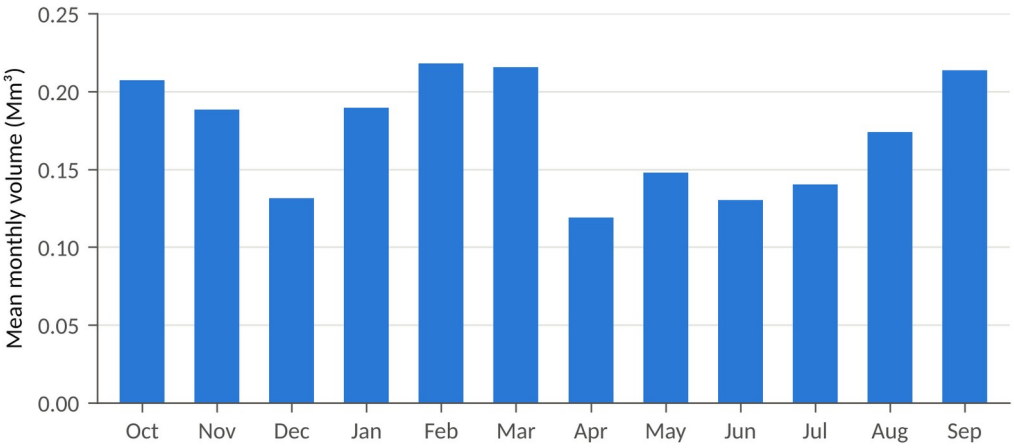


Figure 3: Average monthly volume in the canal from Klein Maricopoort Dam (DWS station A3H031), hydrological years 1997–2014 without 2010.

The inflow of each month is the remainder of the dam's own recorded balance. This remainder is available for 843 months. A further 27 months were rebuilt from the station records and 18 from the WR2012 patched flow, which gives the 888 months from October 1936 to September 2010. Before 1987 a fixed release entry probably overstates the release when the dam was low, and a second, adjusted inflow series was prepared for this period.

No operating rule of the scheme could be obtained. The share of the registered volume that the canal receives when the dam is low was derived from the records, as Coerver *et al*. (2018) do for reservoirs whose rules are unknown. The rule gives this share for seven zones of storage, calculated as the canal volume delivered divided by the volume asked for in all months whose starting storage lay in that zone. The rule was fitted on hydrological years 1966–1974 and 1981–1986 and tested on 1987–2009, the period in which every outflow is measured (Figure 2, lower panel).

5.2.2 Groundwater

A 20 m digital elevation model (DEM) was made in ArcGIS Pro from spot heights and contours. It gives the top of the groundwater model, the ground levels of the boreholes and the levels of the drains.

The model removes water from every cell through a ground drain, which represents seepage, springs, evaporation and uptake by plants. Two depths of the ground drain are used, and they define the two structures of the model. In structure S the ground drain lies 1 m below the land surface. In structure D it lies 15.27 m below the surface, which is the median depth to water of 112 archive levels in A31E and its neighbouring catchments (DWS, s.a.c). These 112 levels include the four calibration levels but none of the 262 levels that are later compared with the model. The depth of 15.27 m is also close to the mean maximum rooting depth of tropical grassland and savanna, 15.0 m (Canadell *et al*., 1996), although such uptake has not been shown in A31E. The base of the weathered zone is taken at 47.5 m below the surface.

The calibration targets are the four static water levels, and the 262 other archive levels are kept for comparison afterwards. The recharge is the regional rate of 29.2 mm/a. The benchmark for the fit is a water table 15.27 m below the ground everywhere.

The abstraction is placed in GIS in three ways. Placement P1 spreads it over the 45 model cells under the land mapped as irrigated, P2 over the 474 cells of surface

deposits within 1 km of that land, and P3 over the 410 bedrock cells within the same distance. Because the effective porosity of A31E is not known, a porosity of 0.10 is used for the particle times, which lies within the range published for fine sand, silt and clay, the surface deposits in which 41 of the 45 cells under the irrigated land lie (Woessner & Poeter, 2020).

5.2.3 Flood

Flood risk was assessed for the reach of the Klein Marico River below the Doringrivier, because the irrigated land lies along it. The reach begins at point F2 and runs 15.7 km downstream (Map 2). The catchment at F2 covers 1 357 km², including the catchment of Klein Maricopoort Dam (F1) and that of the Doringrivier (F3), and its watercourse length and slope were taken from GIS.

The catchment lies in SDF basin 1, with runoff coefficients of 10% for the 1:2-year flood and 40% for the 1:100-year flood, and its time of concentration is 16.4 h (SANRAL, 2013:3-44, 3-46, 3B-2). The design rainfall is the SDF design rainfall over the time of concentration (SANRAL, 2013). The Midgley and Pitman formula was applied as a check with the MAP of 597 mm (SANRAL, 2013:3-61–3-62), and the RMF was applied as an upper limit for Kovács regions K3 and K4, on whose boundary the catchment lies (Kovács, 1988; SANRAL, 2013:3-65–3-67). Each peak was given the triangular hydrograph that the manual prescribes for the SDF, which rises over one time of concentration and falls over two (SANRAL, 2013:3-45, 3-47).

The SRTM model is used for the flood model alone, because the 20 m DEM smoothed the valleys. Manning's n was set for each land-cover group within published ranges (USACE HEC, s.a.b).

5.2.4 Water quality and climate

Surface samples were used for the water-quality screening.

The climate cases are expressed as factors that multiply the value of the Reference run, which is the water balance and the groundwater model with the recorded inflow, evaporation and demand and the regional recharge. Case M represents the median of the projections for ESAF for 2041–2060 under intermediate emissions (SSP2-4.5). Case D represents their dry end, with the 10th percentile of the rainfall and the median

warming, because the projected warming is narrow in range while the rainfall is uncertain even in sign (Section 3.3). The factors are given in Table 5.

Table 5: The climate cases for about 2050, from the projections of the models under SSP2-4.5 for ESAF for 2041–2060 against hydrological years 1937–2009 (calculated from Gutiérrez *et al*., 2021; Iturbide *et al*., 2020). Each factor multiplies the value of the Reference run.

Case	Warming	Annual rain	Inflow	Evaporation and demand	Recharge
Reference	–	–	1.00	1.00	1.00
M (median)	+1.9 °C	−3.1%	0.93	1.064	0.96
D (dry end)	+1.9 °C	−8.5%	0.80	1.064	0.89

The inflow factors follow from the rainfall elasticity of the inflow, which was estimated at 2.0 to 2.5 with the non-parametric estimator of Sankarasubramanian *et al*. (2001:1775). Because the low inflow of 1982 to 1994 was not caused by low rainfall, this elasticity is an approximate scaling rather than a property of the catchments above the dam. The recharge factors follow from the finding that recharge across Africa increases with rainfall to a power of about 1.34 to 1.39 (MacDonald *et al*., 2021). The factor on evaporation and demand corresponds to an increase in potential evapotranspiration of about 3.4% per °C, which lies in the upper half of the “about 1.5–4%” per °C found by Scheff and Frierson (2014:1554), and Cullis and Phillips (2019:25) likewise scale urban water demand by the change in potential evaporation.

5.3 Data processing within software

5.3.1 Dam water balance

The monthly water balance of the dam and its canal was built in Microsoft Excel from the station and reservoir records. It is a mass balance of one reservoir with a restriction rule, which is the form that Adeloye *et al*. (2016) use to calculate reliability, resilience and vulnerability for a single dam. A simpler storage–yield calculation would leave out evaporation, while a river-system model such as WEAP (Dlamini *et al*., 2023) would add components for which no records exist.

In each month, the storage at the end equals the storage at the start, plus the inflow and the rain on the water surface, minus evaporation, the release to the river, the canal delivery and the spill. The balance runs over the 888 months from October 1936 to September 2010 and applies the present capacity of 7.07 Mm³ to every year. It describes the present dam and registered volume under the inflows of the past and is

not a reconstruction of history. The canal receives no water once the storage reaches 0.166 Mm³, which is the lowest storage on record.

5.3.2 Groundwater

The shares of rock group, aquifer type, yield class and soil group, and all elevations, were measured in ArcGIS Pro by overlaying the mapped layers on the boundary of A31E and on the land mapped as irrigated.

The groundwater model was built in GMS 10.9.6 and run with MODFLOW-NWT (Niswonger *et al*., 2011). It consists of one layer, 207.5 m thick, with its top at the 20 m DEM, and 15 142 active cells of about 200 m by 200 m (145 rows and 222 columns). The recharge of 29.2 mm/a (17.53 Mm³/a) is applied to every cell, and the outer edge is closed. Water leaves the model only through drains, which remove water whenever the water level stands above them (Harbaugh, 2005). River drains follow the four named rivers at the level of the lowest land beside the channel, and stream drains follow all other mapped streams at the land surface, both with a conductance of 10 m²/d per metre of channel. The ground drain of every cell has a conductance of 0.05 m²/d per m².

The hydraulic conductivity has two zones, surface deposits (3 479 cells) and bedrock (11 663 cells). It was estimated for each structure with PEST (Doherty & Hunt, 2010) within the limits 0.03–3.0 m/d for the deposits and 0.0005–0.05 m/d for the bedrock, using the four static water levels with an interval of 10 m each. The recharge was held at its regional value, because recharge and conductivity cannot be separated in a calibration on heads alone (Knowling & Werner, 2016). The calibrated model without wells is the base case of each structure.

A31E has no measured aquifer properties for analytical solutions (Leake & Barlow, 2013) and no water-level series for a transient model. The model is therefore run at steady state, in which the water entering the aquifer equals the water leaving it, and the cases are compared with one another. Particle tracking used MODPATH 7 (Pollock, 2016) in the base case of each structure. Nine particles were released at the water table in each of the 45 cells under the irrigated land, 405 in all, and tracked forward.

5.3.3 Flood

The floods were routed with HEC-RAS 7.0.1 in an unsteady two-dimensional model (USACE HEC, s.a.a), in which the water can spread in any direction over the valley floor as the flood passes. The model area is 24.3 km², with a mesh of 30 m cells refined to 15 m along the river. Manning's n, which describes how strongly the land surface resists the flow, was applied by land-cover group. The upstream boundary at F2 is the inflow hydrograph with an energy slope from the terrain, and the downstream boundary is normal depth. The default diffusion-wave equations were used, and the full shallow-water equations as a check.

5.3.4 Water quality and climate

The factors of Table 5 were applied to the inflow, evaporation, demand and recharge of the water balance and the groundwater model described above.

5.4 Analysis

5.4.1 Dam water balance

To pass the test, the simulated monthly storage and yearly canal volume had to meet the criteria of Moriasi *et al*. (2007:891) for a satisfactory simulation, which were set for streamflow and are applied here by analogy. These criteria are a Nash–Sutcliffe efficiency (NSE) above 0.50, a ratio of the root-mean-square error to the standard deviation of the observations (RSR) of 0.70 or less, and a percent bias (PBIAS) within ±25%. The inflow itself cannot be tested, because it is calculated from the storage record. For the same reason, the fit of the monthly storage mainly tests the restriction rule and the outflows, while the yearly canal volume is the more independent test. The test is independent only for the restriction rule and the demand, because the area curve, the pan factors and the release table were derived from the test period itself, and passing a split test is in any case "a necessary, rather than a sufficient, condition" for a model to be adequate (Klemeš, 1986:13). After the test, the rule was refitted on both periods together, so that the rule used in all further runs also contains the test period. Its shares rise from 0.036 in the lowest zone to 1.00 above 57% of capacity. The second, adjusted inflow series was run as a check.

The supply is judged by the criteria of Hashimoto *et al*. (1982), calculated with the formulas of Adeloye *et al*. (2016). A month fails when the delivery is below 99% of the

registered volume of that month. Reliability by time is the share of months without failure, reliability by volume the share of the registered volume delivered, and yearly reliability the share of years with the full volume. Resilience is the number of shortfall events, which are unbroken runs of failure months, divided by the number of failure months, and vulnerability is the mean depth of a shortfall. The shortfall of a month is the registered volume less the simulated delivery at the head of the canal, and it is referred to as the shortfall against the registered volume throughout. Every criterion is a frequency counted in one record of 74 years and not a probability, since synthetic series (Kjeldsen & Rosbjerg, 2004) would require a stochastic model of the inflow. A curve of reliability against demand was run for demands of 0.5 to 3.0 Mm^3/a. The historic firm yield is the largest demand that the dam could have met in every month of the record without restriction.

The records cannot distinguish a month in which the canal was restricted from a month in which less water was wanted, and every run is made twice. With the restriction rule, the canal receives the share given by the rule, while without restriction it receives the full volume for as long as water remains in the dam. The unrestricted run gives the upper bound of reliability, because every restricted month counts as a failure and much of the water held back in the dam is lost to evaporation. The run with the rule gives the lower bound for the scheme as it was operated on average over the record, but it is not the lowest possible value, since a rule that restricts more would lower the reliability further (Section 6.3). For resilience and vulnerability the order is reversed, because a restricted canal fails more often but less deeply, and the two bounds are always reported together. The sensitivity of the results was tested by changing one input at a time (Pianosi *et al*., 2016:216), namely the inflow series, evaporation, the capacity, the release to the river, the rule and the seasonal pattern of demand.

5.4.2 Groundwater

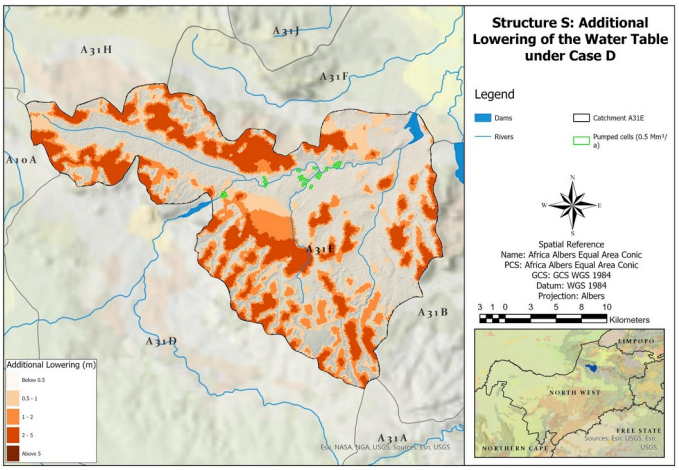
The shortfall was first compared with the groundwater balance of the Reserve determination (DWS, 2021). Such a balance treats recharge as water that is free to be taken, while pumped water is balanced mostly by capture (Section 3.2), and only a flow model can show which discharge is reduced by pumping. With four static water levels and a steady-state calibration, the model belongs to Class 1, the lowest class of confidence in the Australian groundwater modelling guidelines, whose uses include

developing coarse relationships between abstraction and impact (Barnett *et al*., 2012:20–21). Both structures are needed because the records contradict each of them (Section 6.4) and cannot identify a single conceptual model (Section 3.2). The fit was compared with the benchmark, and the 262 other archive levels were compared with both structures afterwards. Neither comparison is a validation (SKM, 2013:18).

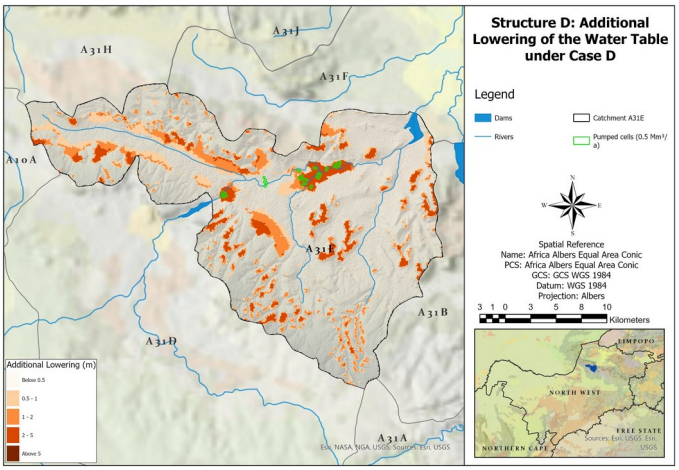
Abstraction was run at rates of 0.25 to 2.34 Mm³/a at placement P1. Placements P2 and P3 test whether spreading the abstraction changes the result. The recharge was also multiplied by the factors of cases M and D, 0.96 and 0.89, both without wells and with 0.5 Mm³/a at P1.

Four results are read from each case and compared with the base case at the same recharge. These are the rate actually pumped, because MODFLOW-NWT reduces the rate of a well when its cell nearly dries (Niswonger *et al*., 2011), the drawdown in the pumped cells, the number of pumped cells in which the water level falls below the base of the weathered zone at 47.5 m, and the change in the outflow of each group of drains as a share of the rate pumped. A reduction in the river or stream drains is water that would otherwise have flowed to Kromellenboog Dam. The additional lowering of the water table caused by a lower recharge is read from two runs that both include the abstraction, one at the base recharge and one at the reduced recharge (Map 7).

(a)



(b)



Map 7: Additional lowering of the simulated water table when recharge falls to 89% of the base value (case D), for (a) structure S and (b) structure D, with 0.5 Mm³/a abstracted from the 45 model cells under the land mapped as irrigated in both runs.

A steady-state drawdown is the state reached if pumping continued indefinitely at the mean recharge. It exceeds the drawdown of a single season of pumping at the same recharge, but it is not an upper limit for a long drought, during which the recharge falls below its mean. The loss of streamflow would in reality arrive later and continue after pumping stops (Section 3.2). Two further runs show what the four water levels cannot decide. In one, the recharge and conductivities are halved or doubled, which fits the levels equally well, and in the other structure D is run with the conductivities of structure S.

The particle times are those of the moving water alone, without dispersion, retention or decay. A porosity of 0.01, within the range for fractured crystalline rock, would make the times ten times shorter than those at 0.10.

5.4.3 Flood

As the drainage manual requires, the SDF was used as the main estimate and compared with the other applicable methods (SANRAL, 2013:3-4). Because the manual warns that the flood of a dolomitic catchment may be less than half of the SDF (SANRAL, 2013:3-45), the SDF is treated as the conservative case. Four plans were run, namely the 1:50-year and 1:100-year SDF floods, the 1:100-year Midgley and Pitman flood, and the 1:100-year SDF flood with the full shallow-water equations as a check on the default diffusion-wave equations. Two simplifications act in opposite directions, since ignoring the attenuation in Klein Maricopoort Dam makes the flood larger, while leaving out the inflow of the Waterkloofspruit makes the flood below its junction smaller.

5.4.4 Water quality and climate

Each water-quality quantity was compared with its guideline value, and the share of samples above the guideline was counted for each site. A value above a guideline shows that the water could affect its use or its ecosystem, but it does not show that the water is contaminated, and the shares of single samples are a screening indication and not a trophic classification (DWAF, 1996b:84). The river below the upper dam and Kromellenboog Dam were also compared over their common period, 2012–2023, with a simple mixing check based on the share of the lower dam's inflow that comes from the upper dam.

In the climate cases, the dam was run with the inflow multiplied by 0.93 and 0.80 and evaporation and demand by 1.064, and the reliability is measured against the registered volume raised by this factor. The groundwater model was run with the recharge multiplied by 0.96 and 0.89 (Section 5.4.2).

6. Results

6.1 Setting of the irrigated land and the records

A31E falls from 1 567 m in the south to 1 038 m at Kromellenboog Dam (Map 2). Between the two dams, the river falls about 104 m over a distance of 24.7 km. The Doringrivier joins the Klein Marico River 4.2 km below the dam wall and the Waterkloofspruit 13.7 km below it. The water surface of the upper dam stands at about

1 150 m, while all the land mapped as irrigated below it lies lower, between 1 093 and 1 140 m, so that a canal leaving the dam could reach this land by gravity.

Quartzite and sandstone cover 31.8% of A31E, the andesitic lava of the Hekpoort Formation and diabase 24.4%, mudrock, shale and hornfels 20.9%, and surface deposits 22.5%. On the hydrogeological map, 59.8% of A31E is classified as an intergranular and fractured aquifer and 40.2% as a fractured aquifer. Of the land mapped as irrigated, 91.6% lies on surface deposits. The thickness and composition of these deposits are not known. The mapped median borehole yield is 0.5–2.0 L/s over 75.3% of A31E and 0.1–0.5 L/s over the remaining area, which includes three-quarters of the irrigated land. The registered scheme volume is equal to about 74 L/s pumped continuously, which would require about 150 to 740 boreholes at yields of 0.5 to 0.1 L/s. According to the modelled regional values of the GRAII (Section 2.4), the weathered zone holds about 91% of the stored groundwater.

Each of the 266 boreholes with a water level has a single level. The 262 levels that are not recorded as static have a median depth of 19.4 m, and the median strike yield of 109 boreholes is 0.3 L/s.

The soils of A31E belong to the hydrological soil groups B/C (38.7%), C (31.5%), C/D (28.7%) and B (1.2%). The soils with the highest runoff potential lie along the river valley, and 83.2% of the land mapped as irrigated lies on group B/C. The Wilgeboomspruit yielded about 53 mm/a, or 7.6% of the rainfall, in 1966–1980 (DWS, s.a.b). Natural woodland and grassland cover 87% of A31E and cultivated land 10.2%, more than half of which is fallow. Land mapped as irrigated covers 168 ha, of which 162 ha lie in fifteen patches in the strip below the dam (Map 6). Built-up land covers 0.3%.

Half of the scheme positions of the register lie on higher ground above the level of the dam, which a canal cannot reach by gravity. Three different quantities describe irrigation from the dam. The register lists 442 ha and 2.34 Mm³/a, the land cover maps 168 ha of irrigated land, of which 145 ha is under centre pivots, and the canal carried 1.89 to 2.19 Mm³/a in 2022–2024 (DWS, s.a.b).

The yearly rainfall at the dam varied between 314 and 900 mm in 1967–2008 (Figure 4). Every year from 1981 to 1986 fell below the long-term mean, which is the longest such run since 1920 (Bailey & Pitman, 2016; DWS, s.a.b). At the dam, 93% of

the rain falls between October and April, and the S-pan evaporation averages 2 005 mm/a and exceeds the rainfall in every month (Figure 5; DWS, s.a.b). The reservoir loses a mean of 1.07 Mm³/a to evaporation over the record of 1936–2010, and about 2.0 Mm³/a when the dam is nearly full (DWA, 2012).

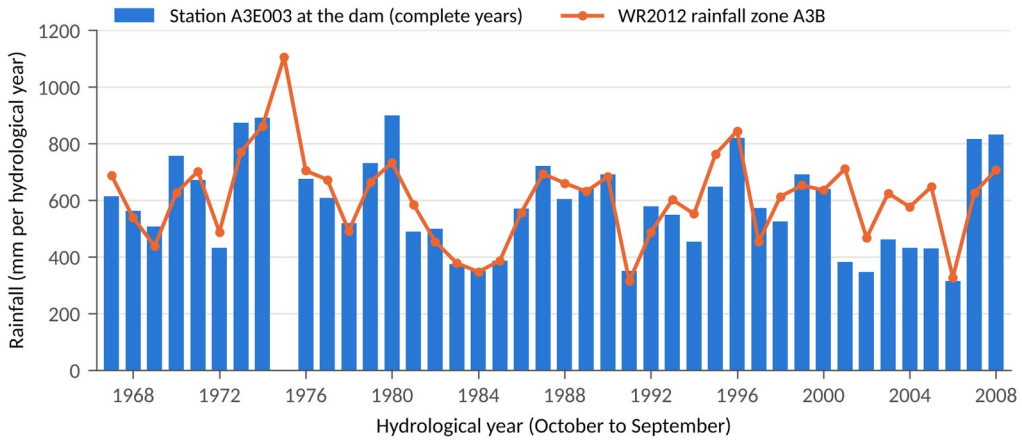


Figure 4: Yearly rainfall at Klein Maricopoort Dam (DWS station A3E003, complete hydrological years only) and in WR2012 rainfall zone A3B, hydrological years 1967–2008. Hydrological year 1975 is left out, because its total includes an implausible 670 mm in August 1976.

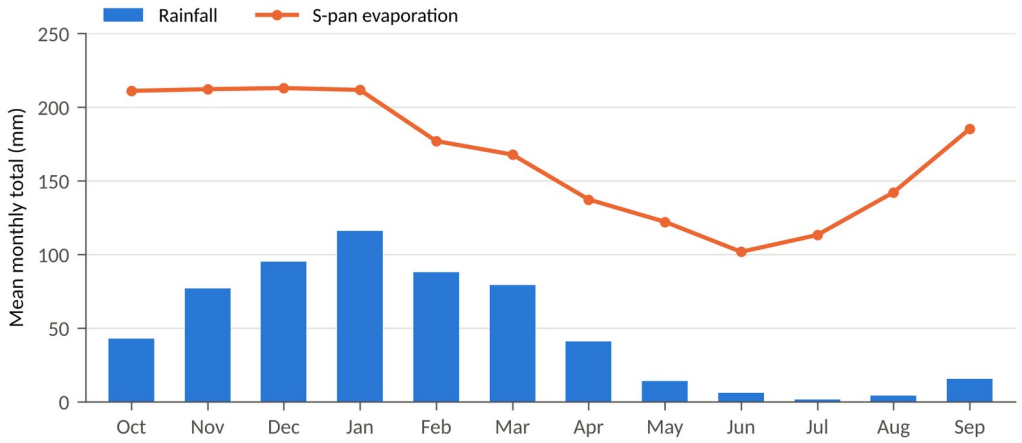


Figure 5: Average monthly rainfall and S-pan evaporation at Klein Maricopoort Dam (DWS station A3E003), hydrological years 1967–2014.

6.2 The canal and the water of A31E

The inflow to Klein Maricopoort Dam, taken as the WR2012 patched observed flow, has a median of 3.94 Mm³/a over the hydrological years 1936–2009 (Figure 6). It stayed below this median from 1982 to 1994, although the rainfall had recovered by 1987, and over the ten years from 1985 to 1994 it averaged 0.84 Mm³/a, which is about a third of the registered volume before any evaporation. Over 1987–2009 the inflow from the river averaged 3.33 Mm³/a (Table 6). Evaporation took a third of all the water that entered the dam over 1987–2009 and about half in 1987–1995, when the canal carried 0.30 Mm³/a. The survey capacity of 7.07 Mm³ equals about three years of the

registered scheme volume, and about 1.9 years once evaporation and the release to the river are included. At the start of the drought, inflows of 2.14 and 0.59 Mm³ in 1982 and 1983 lowered the dam from 99% of its capacity in December 1982 to 22% in March 1984, and it stayed low until the rains of 1995 and 1996 (DWA, 2012).

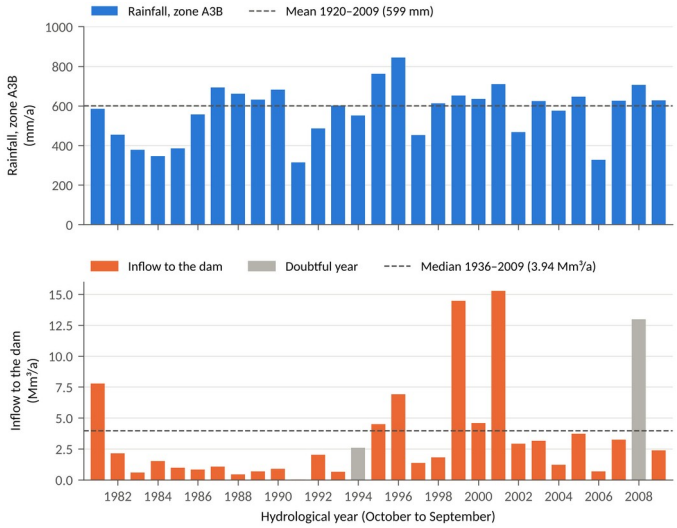


Figure 6: Yearly rainfall in WR2012 rainfall zone A3B (upper panel) and yearly inflow to Klein Maricopoort Dam as WR2012 patched observed flow (lower panel), hydrological years 1981–2009, with the long-term mean rainfall (1920–2009) and the median inflow (1936–2009) (Bailey & Pitman, 2016; DWA, 2012).

Table 6: Water balance of Klein Maricopoort Dam from the records, in Mm³/a (DWA, 2012; DWS, s.a.b). The inflow is the remainder that closes the balance. The drought column runs from 1987, the first year in which every outflow is gauged, to 1995, the hydrological year in which the dam recovered.

Term	1987–2009	1987–1995 (drought)	1996–2009
Inflow from the river (remainder)	3.33	1.23	4.67
Rain on the water surface	0.42	0.24	0.53
Evaporation	1.23	0.74	1.55
Canal	1.20	0.30	1.78
Spill	0.87	0.00	1.43
Release to the river	0.33	0.07	0.49
Change in storage	+0.11	+0.36	−0.05

Figure 7 shows the storage of the dam and the yearly canal volumes. Of the 38 countable years of the raised dam, the canal carried less than the registered volume in 24 years (63%), with a mean shortfall in those years of 58% of the registered volume. The longest run of shortfall years lasted ten years, from 1984 to 1993 (DWS, s.a.b). In five of the shortfall years the dam never fell below 75% of its capacity.

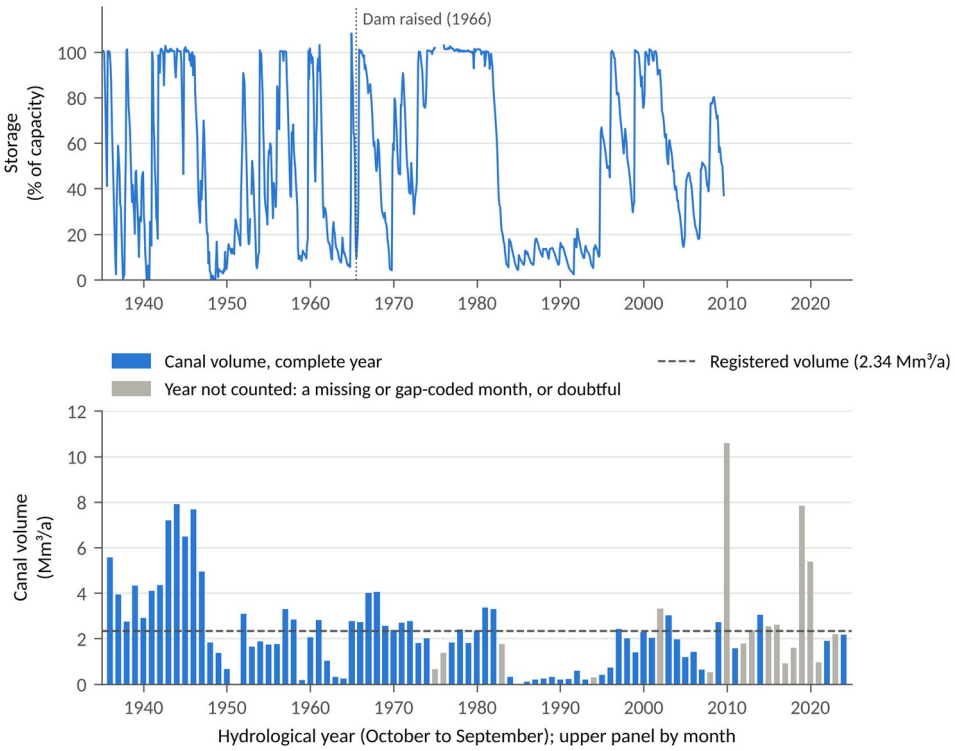


Figure 7: Upper panel: monthly storage in Klein Maricopoort Dam as a percentage of the capacity at the time, June 1936 to November 2010 (DWA, 2012). Lower panel: yearly volume in the canal, hydrological years 1936–2024 (DWS station A3H031), with the registered scheme volume of 2.34 Mm³/a. Years that are not counted are drawn in grey.

The water of A31E behaved differently. Over 1987–2009, 5.01 of the 6.21 Mm³/a (81%) that reached Kromellenboog Dam did not come from the upper dam (DWA, 2012; DWS, s.a.b). In 1987–1995, the net contribution of the reach between the dams was two-thirds of its volume in 1996–2009, 3.80 against 5.80 Mm³/a, under almost the same rainfall, while the inflow from upstream fell to a quarter. The reach supplied 98% of the inflow to Kromellenboog Dam in these years. The net contribution is positive in every year and has a rank correlation of 0.76 with the zone rainfall. The modelled naturalised runoff shows no such reduction and is about twice the net contribution during the drought. The Wilgeboomspruit, which had flowed in every dry-season month for eighteen years and yielded a mean of 1.27 Mm³/a in 1966–1990, was dry in 42 of the 50 months from February 1984, although monthly rains above 100 mm continued. It flowed again from April 1988, but at a third of its former mean (Figure 8).

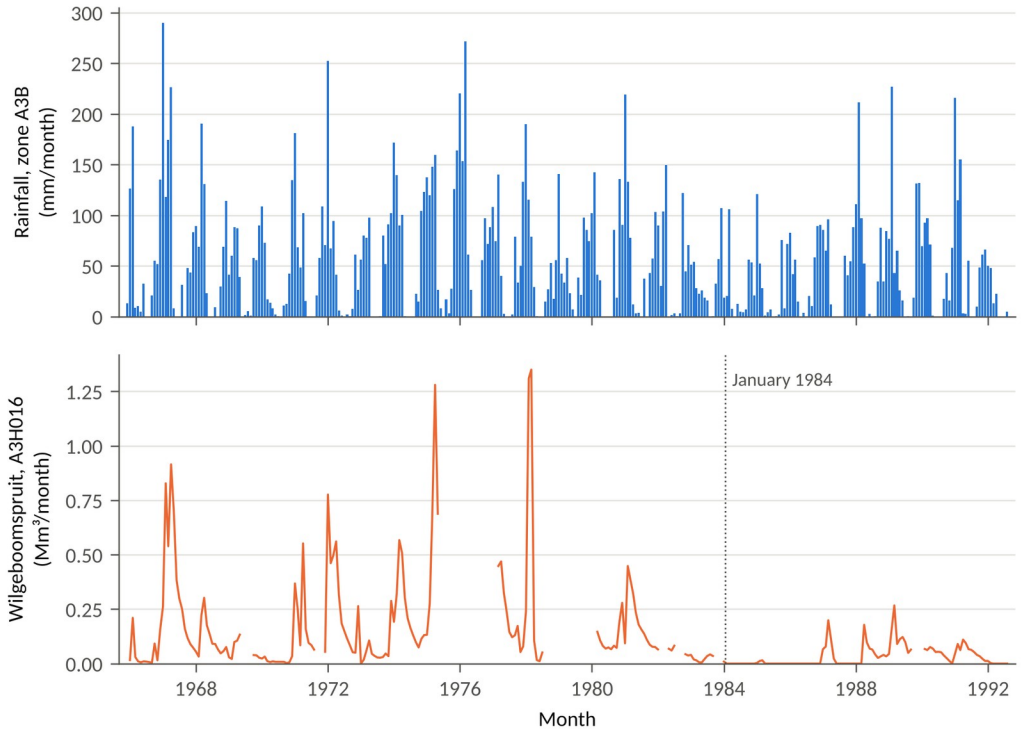


Figure 8: Monthly rainfall in WR2012 rainfall zone A3B (upper panel) and monthly flow of the Wilgeboomspruit at DWS station A3H016 (lower panel), December 1965 to August 1992.

6.3 Reliability of the canal supply

In the test period of 1987–2009, the simulated monthly storage meets all three criteria (NSE 0.86, RSR 0.37, PBIAS +7.4%), and so does the yearly canal volume (NSE 0.60, RSR 0.64, PBIAS −22.4%) (Figure 9). The balance delivers 22% more water than the canal carried, mainly in the years of recovery after the drought, when the canal took up water more slowly than the dam could supply it. The monthly canal volume does not meet the criteria, although the balance finds a shortfall in 192 of the 265 months against 216 in the record.

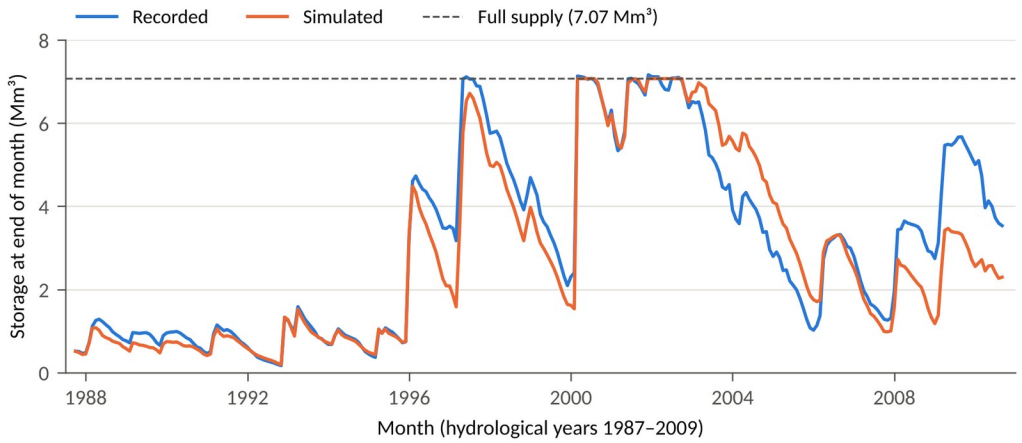


Figure 9: Simulated and recorded monthly storage of Klein Maricopoort Dam, hydrological years 1987–2009 (test of the restriction rule; recorded storage from DWA, 2012).

Over the 74 years, the canal receives the full registered volume in 59% to 87% of the months, and a year passes without a shortfall in 45% to 77% of the years (Table 7). With the rule the canal fails three times as often, but a failure month still brings half of the registered volume, while without restriction the canal then receives almost nothing. One drought dominates every count (Figure 10). With the rule, it forms a single event of 146 months, from December 1983 to January 1996, with a total shortfall of 21.9 Mm³, which is more than nine times the yearly volume and includes 40% of all failure months. Without restriction, the same drought breaks into separate events of 11 to 20 months, because the inflow of each rainy season is passed to the canal at once.

Table 7: Reliability, resilience and vulnerability of the canal supply against the registered volume of 2.34 Mm³/a, Reference run, hydrological years 1936–2009.

Criterion	With the restriction rule (lower bound)	Without restriction (upper bound)
Reliability by time	0.590	0.867
Reliability by volume	0.795	0.883
Yearly reliability	0.446	0.770
Shortfall events; longest (months)	16; 146	11; 20
Resilience	0.044	0.093
Vulnerability	0.50	0.88
Mean shortfall (Mm³/a)	0.48	0.27

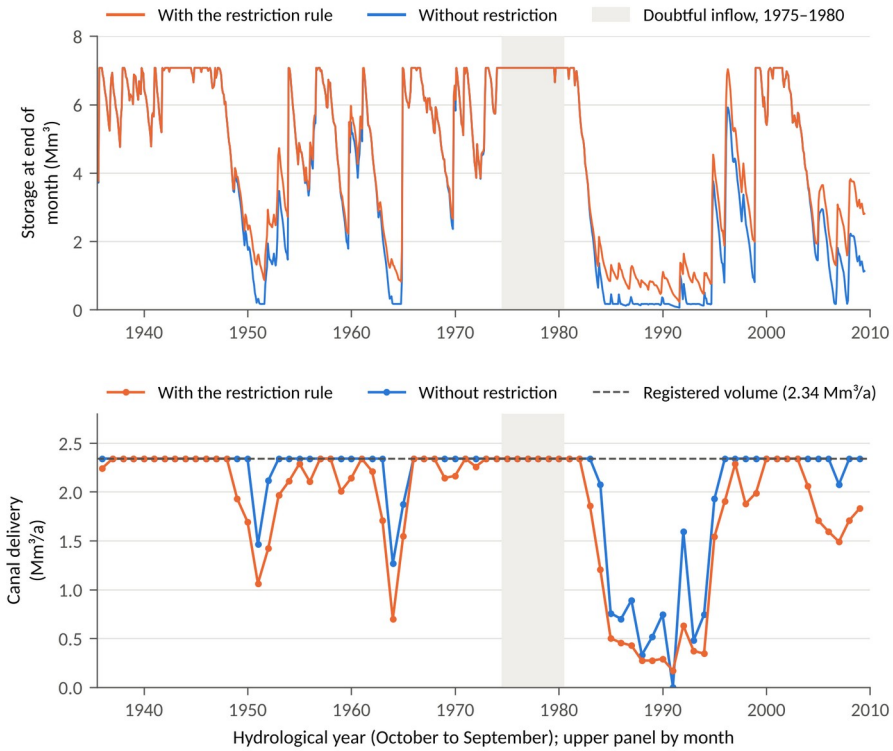


Figure 10: Reference run of the water balance of Klein Maricopoort Dam, hydrological years 1936–2009, with the restriction rule and without restriction. Upper panel: simulated monthly storage. Lower panel: yearly canal delivery against the registered volume of 2.34 Mm³/a. Shaded: the years 1975–1980, in which the inflow is doubtful.

The answer depends more on the restriction rule than on the demand (Figure 11). Without restriction, a lower demand helps, and the largest demand that the dam could have met in every month of the record is about 0.36 Mm³/a, since one month already fails at 0.38 Mm³/a. This historic firm yield is about a quarter of the 1.4 Mm³/a that the Department gives as the yield of the dam (DWAF, 2004:50), and at 1.5 Mm³/a the unrestricted canal fails in 8% of the months. With the rule, a firm yield cannot be defined, because the rule restricts the canal whenever the dam falls into its low zones. Even at a demand of 0.5 Mm³/a, the rule leaves a reliability by time of 0.82 and an event of 128 months. The difference between the two bounds at the registered volume, 0.28 in reliability by time, is about as large as the effect of raising the demand from 0.5 to 3.0 Mm³/a.

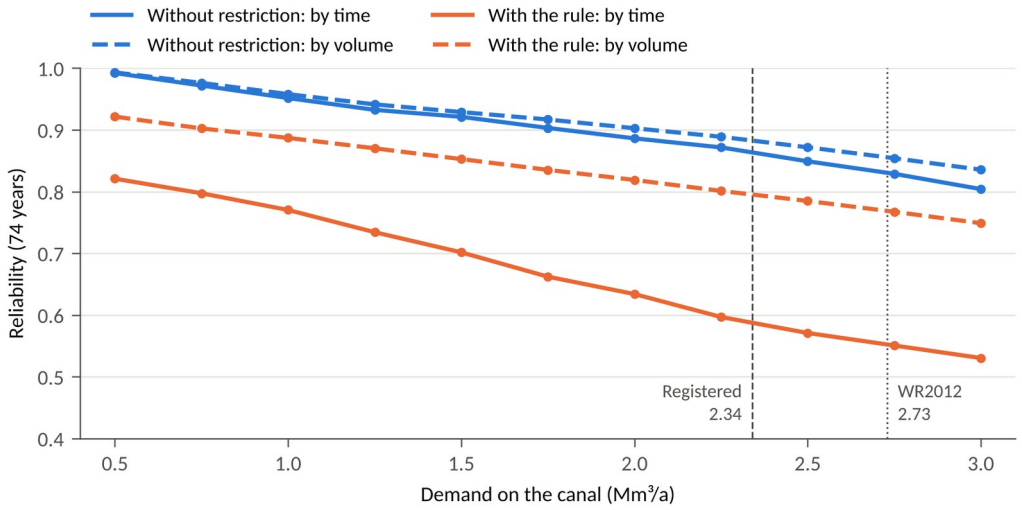


Figure 11: Reliability of the canal supply by time and by volume against the demand on the canal, 0.5 to 3.0 Mm³/a, with the restriction rule and without restriction, hydrological years 1936–2009. Vertical lines: the registered volume (2.34 Mm³/a) and the demand of the WR2012 study (2.73 Mm³/a).

The rule itself is the most sensitive input. The rule of the test period alone, which restricts more than the rule used, lowers the reliability by time from 0.59 to 0.39, and no run with a rule comes near the 0.87 of the unrestricted canal. The other inputs matter far less. With the rule, a change of 10% in evaporation or a change in the release to the river each moves the reliability by time by about 0.02, and every other input, including a capacity 10% smaller and the seasonal pattern of demand, moves it by less than 0.01. Without restriction, the largest change, also about 0.02, comes from a higher lowest usable storage. Leaving out the doubtful years lowers the reliabilities by at most 0.05.

6.4 Groundwater model: fit and water budget

For structure S, the calibration gave conductivities of 0.430 m/d for the surface deposits and 0.0029 m/d for the bedrock. Structure S matches three of the four static levels within 6 m, but it misses borehole 2526AD00180 in the valley by 28.9 m. Its root mean square error over the four levels is 14.9 m, against 8.5 m for structure D and 8.6 m for the benchmark. For structure D, both conductivities ended at the lower limits of their ranges, 0.03 and 0.0005 m/d. With the regional recharge, the levels are matched only by a bedrock transmissivity of 0.60 m²/d in structure S and 0.10 m²/d in structure D, which is far below the regional 22.81 m²/d.

The 262 archive levels that were not used in the calibration are compared with both structures in Figure 12. In structure S, 31% of these levels lie within 10 m of the model, and where the ground lies below 1 100 m, the measured water table is on average 23.1 m deeper than the modelled one. Nine of the recorded levels are shallower than 5 m, while structure S places the water table this shallow at 114 boreholes. Structure D halves the misfit on the low ground, and 62% of the levels lie within 10 m. On these 262 levels it fits no better than the benchmark, with a root mean square error of 16.9 m against 14.9 m, compared with 24.3 m for structure S. Under the irrigated land, the water table lies 4.5 m deep in structure S and 15.3 m deep in structure D (Map 8).

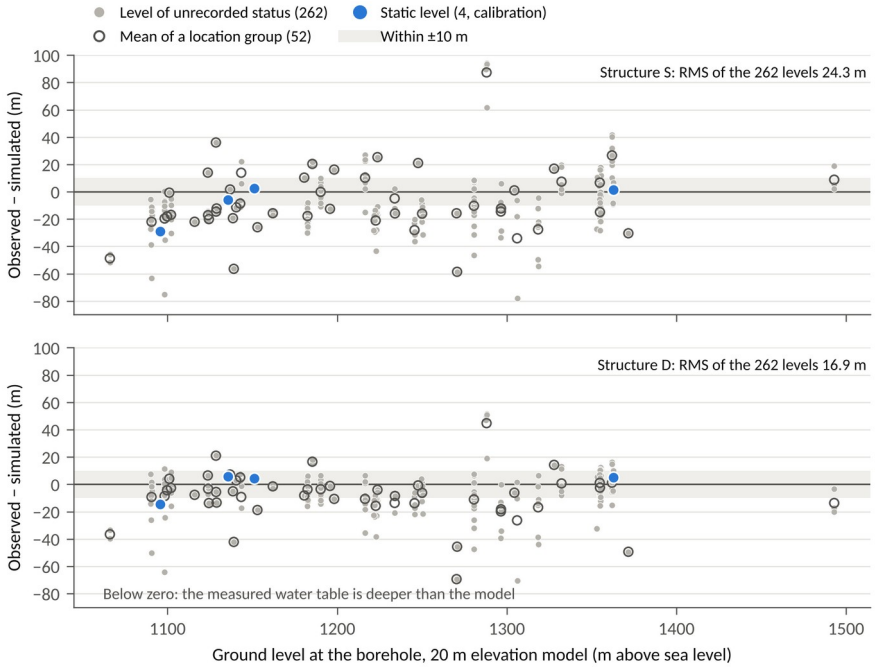
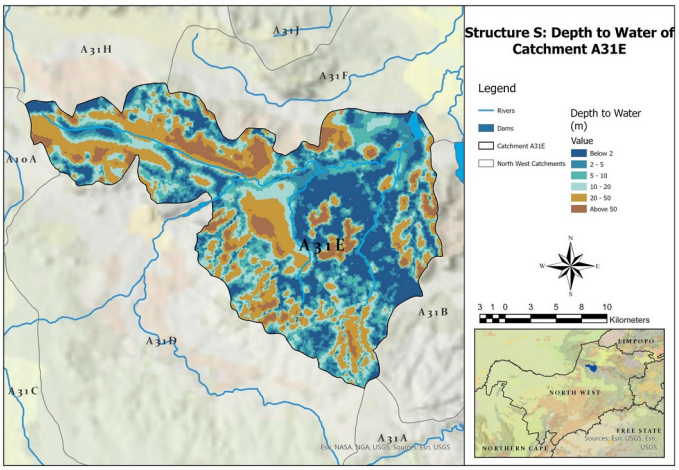
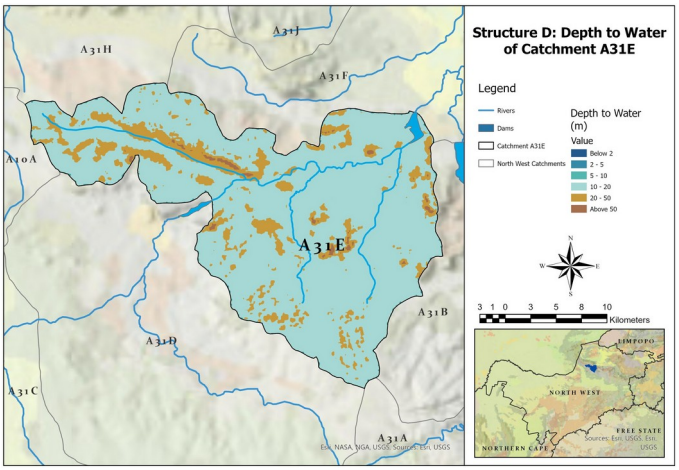


Figure 12: Residual (observed minus simulated head) at the single water levels recorded inside A31E (National Groundwater Archive, 1910–1994) against ground level, for structure S (upper panel) and structure D (lower panel). Grey: levels whose status is not recorded; open circles: the mean of each location group; blue: the four static levels.

(a)



(b)



Map 8: Simulated depth to water in the groundwater model of A31E for (a) structure S (outflow 1 m below the surface) and (b) structure D (outflow 15.27 m below the surface), with the rivers and streams and the two dams.

The two structures receive the same recharge but dispose of it in opposite ways (Table 8). In structure S, more than half of the recharge, 9.86 Mm³/a, reaches the rivers and streams. This exceeds every estimate of baseflow, which ranges from 2.00 Mm³/a in the Reserve determination (DWS, 2021) to 3.43 Mm³/a by the method of Hughes in the WR2012 study (Bailey & Pitman, 2016), and it exceeds the net contribution of the reach of 5.01 Mm³/a. In structure D, almost all the recharge leaves through the ground drain, and the rivers and streams receive 0.63 Mm³/a, which is below every estimate. Halving or doubling the recharge and conductivities together leaves the heads unchanged and scales these flows in proportion. No such model of structure S falls within the range of the baseflow estimates (4.93 Mm³/a at half), and structure D remains below it (at most 1.26 Mm³/a).

Table 8: Water budget of the two base cases of the groundwater model, without wells, in Mm³/a. The model applies the regional recharge rate of 29.2 mm/a to every cell, which gives slightly more than the 17.336 Mm³/a of the Reserve determination. The budget closes to 0.00% in both structures.

Term	Structure S	Structure D
In: recharge	17.53	17.53
Out: ground drain	7.67 (44%)	16.90 (96%)
Out: rivers and streams	9.86 (56%)	0.63 (4%)

The Reserve determination gives a recharge of 17.336 Mm³/a, a groundwater use of 0.81 Mm³/a against the 0.285 Mm³/a registered (Section 1.1), and a discharge to the streams of 2.00 Mm³/a (DWS, 2021). The remaining 14.5 Mm³/a is unaccounted for. The recharge is five to nine times the estimates of baseflow. The national estimates are of the same size, at 4.84% to 4.89% of the rainfall, and the GRAII gave about 56 mm/a before its final calibration (DWS, s.a.a).

6.5 Abstraction to offset the shortfall

In structure S, the mean drawdown in the 45 pumped cells increases from 0.96 m at 0.25 Mm³/a and 2.30 m at 0.5 Mm³/a to 40.8 m at the registered volume of 2.34 Mm³/a (Figure 13). At this rate the water level in 19 of the cells lies below 47.5 m, the base of the weathered zone. In structure D, the drawdown is five and nine times larger, at 4.92 m for 0.25 Mm³/a and 20.6 m for 0.5 Mm³/a (lower limits, Section 6.4), and from 1.0 Mm³/a onwards the cells cannot deliver what is asked, as they yield 0.93 Mm³/a at 1.0 Mm³/a and at most about 1.2 Mm³/a. The source of the pumped water is also opposite in the two structures. In structure S, 87% to 89% of the pumped water is balanced by less discharge to the rivers at rates of up to 1.0 Mm³/a, and still 78% at the registered volume. At 0.5 Mm³/a, about 69% comes from the Klein Marico River itself and 20% from the lower Waterkloofspruit. In structure D, the rivers and streams supply 0.3% to 1.5%, and the rest comes from the deep outflow.

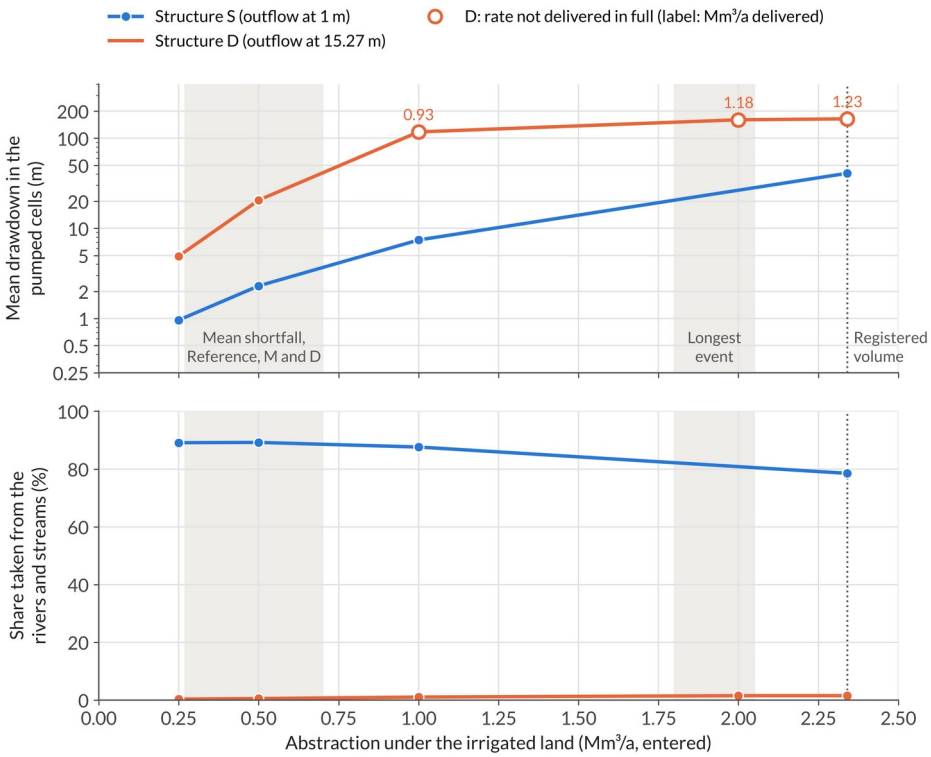


Figure 13: Steady-state response of the groundwater model to abstraction spread over the 45 model cells under the land mapped as irrigated, for structure S and structure D. Upper panel: mean drawdown in the pumped cells; lower panel: share of the abstraction balanced by less discharge to the rivers and streams. Open circles: rates that structure D cannot deliver. Shaded: the mean shortfall against the registered volume in the Reference run and cases M and D, for both bounds (0.27–0.70 Mm³/a), and the rate during the longest shortfall event of the Reference run (1.80–2.05 Mm³/a).

Table 9 lists the cases that change one thing. With the conductivities of structure S, structure D lowers the water level by 2.22 m, which is about as much as structure S, and its water still comes entirely from the ground drain. Spreading the abstraction over the surface deposits within 1 km reduces the drawdown to a quarter in structure S and to a twenty-eighth in structure D, without changing the source. Models with half or double the recharge and conductivities, which fit the four water levels equally well, give drawdowns of 0.96 to 7.43 m in structure S. In structure D they give 4.92 m and 117.5 m, and the halved model cannot deliver the abstraction.

Table 9: Mean drawdown in the pumped cells, and share of the abstraction balanced by less discharge to the rivers and streams, for 0.5 Mm³/a in the cases that change one thing. The halved and doubled models give the same heads as the calibrated model at 1.0 and 0.25 Mm³/a, so their values equal those runs (Section 5.4.2).

Case	S: drawdown (m)	S: from rivers and streams	D: drawdown (m)	D: from rivers and streams
P1, under the irrigated land	2.30	89.2%	20.6	0.5%
P2, spread over the deposits within 1 km	0.55	88.8%	0.73	2.7%
P3, on the bedrock within 1 km	3.93	58.0%	Not run	Not run
P1, recharge × 0.89 (case D)	2.42	89.8%	24.3	0.6%
P1, conductivities of S in structure D	–	–	2.22	0%
P1, recharge and conductivities halved	7.43	87.6%	117.5 (not delivered)	1.0%
P1, recharge and conductivities doubled	0.96	89.1%	4.92	0.3%

The mean shortfall of the Reference run, 0.27 to 0.48 Mm³/a, is 1.6% to 2.8% of the modelled recharge, and the registered volume is 13.5%. The mean shortfall equals the continuous yield of 29 to 51 boreholes at the median strike yield of 0.3 L/s. The rate needed during the longest shortfall event, 1.8 to 2.05 Mm³/a, cannot be delivered from under the irrigated land in structure D, and in structure S it lies between runs with drawdowns of 7 and 41 m.

6.6 Climate change

Table 10 compares cases M and D with the Reference run. At the median of the projections, the reliability by time decreases by 5 percentage points with the rule and by 4 points without restriction, while at the dry end it decreases by 11 and 8 points. Without restriction, the scheme would receive its full yearly volume in about seven out of ten years in case M and in fewer than two out of three years in case D, compared with three out of four years in the Reference run. The mean shortfall increases by a quarter to a third in case M and by a half to two-thirds in case D. Cases M and D share the same warming and differ in inflow and recharge, but with the rule case D lies 6 points below case M, which is more than case M lies below the Reference run. In case D, the longest event also grows from 146 to 163 months, and the months with less than half the registered volume increase from 156 to 214 with the rule.

Table 10: The climate cases against the Reference run. Canal supply: with the restriction rule / without restriction, hydrological years 1936–2009. Groundwater: discharge to rivers and streams in the base case, and its change under the recharge factor of the case.

Case	Reliability by time	Yearly reliability	Mean shortfall (Mm³/a)	Longest event (months)	Groundwater to rivers, S	Groundwater to rivers, D
Reference	0.590 / 0.867	0.446 / 0.770	0.48 / 0.27	146 / 20	9.86 Mm³/a	0.63 Mm³/a
M (median)	0.541 / 0.829	0.405 / 0.703	0.60 / 0.36	148 / 35	−3.8%	−5%
D (dry end)	0.477 / 0.791	0.324 / 0.635	0.70 / 0.44	163 / 35	−10.5%	−14%

Cases M and D also lower the storage of the dam at all times and not only during droughts (Figure 14). With the rule, the dam holds at least half of its capacity in about 64% of the months in the Reference run, compared with about 59% in case M and 53% in case D.

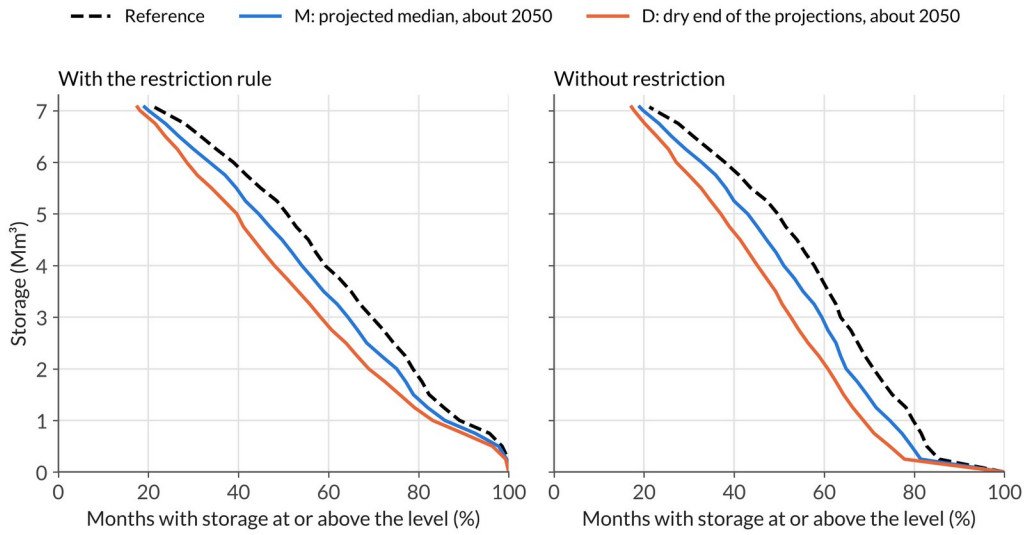


Figure 14: Share of months in which the simulated storage of Klein Maricopoort Dam is at or above each level, for the Reference run and cases M and D, hydrological years 1936–2009. Left panel: with the restriction rule; right panel: without restriction.

In the groundwater model, a lower recharge reduces the discharge to the rivers and streams by about as much as the recharge. The discharge decreases by 3.8% and 10.5% in structure S, which is slightly less than the recharge, and by 5% and 14% in structure D, which is more than the recharge. Map 7 shows the additional lowering of the water table under case D, from two runs that both include an abstraction of 0.5 Mm³/a under the irrigated land, so that the drawdown of the abstraction itself is removed. In structure S, the additional lowering has a mean of 1.00 m and exceeds 1 m over 220 of the 602 km², mainly on the high ground between the valleys. In the valleys the drains hold the water table, and under the pumped cells the lowering averages 0.23 m. In structure D, the lowering is small over most of the catchment, with a mean of 0.38 m, but it is largest under the pumped cells, where it averages 3.78 m

and the drawdown of the abstraction increases from 20.6 to 24.3 m. Under case M, the additional lowering is about a third of that under case D.

6.7 Flood risk in the modelled reach

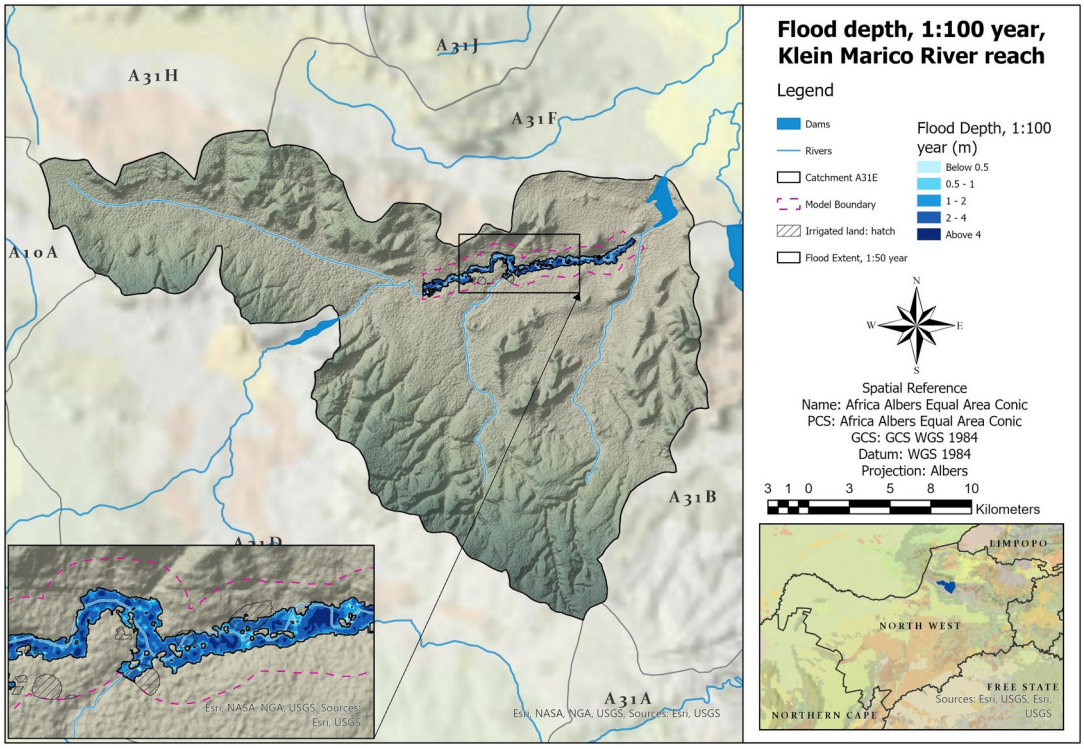
The catchment above the reach is underlain for 46.4% by dolomite of the Malmani Subgroup, all of which lies above Klein Maricopoort Dam, while A31E itself contains none (Michaluk & Keyser, 1991; Walraven, 1981). The model area contains 8.04 km² of land mapped as cultivated (DFFE, 2024).

The three methods give design floods that differ widely in size (Table 11). At the 1:100-year return period, the SDF peak of 1 252 m³/s exceeds the Midgley and Pitman peak by 41%, and it also exceeds the RMF of region K3. The Midgley and Pitman peak is 71% of the SDF peak.

Table 11: Design flood peaks at F2 and the floods routed through the modelled reach (catchment about 1 357 km²; model area 24.3 km², with 8.04 km² of land mapped as cultivated). RMF (no return period): 1 205 m³/s for region K3 and 2 355 m³/s for region K4.

Plan	Design peak (m³/s)	Peak in / out (m³/s)	Peak reduced	Flooded (km²)	Deeper than 1 m (km²)	Cultivated land flooded (km²)
1:50, SDF	974	972 / 951	2.1%	7.72	6.26	3.17
1:100, SDF	1 252	1 250 / 1 236	1.1%	7.97	6.66	3.24
1:100, Midgley and Pitman	889	887 / 858	3.3%	7.60	6.10	3.13
1:100, SDF, shallow-water equations	1 252	1 250 / 1 234	1.3%	8.34	7.01	3.33

Over the 15.7 km of the reach, the peak decreases by 1% to 3% and arrives about two hours later, because the floodplain holds at most 14% to 16% of the flood volume. The flooded area changes far less than the peak. The Midgley and Pitman flood has a peak 29% lower than the SDF flood, but it covers 5% less land. The outlines of the 1:50-year and 1:100-year SDF floods almost coincide on Map 9. The 1:100-year flood covers 0.25 km² more land, which along the two edges of the 15.7 km reach moves the edge outwards by about 8 m on average, less than a third of one 30 m cell of the terrain. The full shallow-water equations change the flooded area by 5% and the outflow peak by 0.2%. In every plan, 3.1 to 3.3 km² of land mapped as cultivated is flooded, which is about 40% of the cultivated land in the model area, and most of it lies under more than 1 m of water. This includes about 44 ha of the land mapped as irrigated under the 1:100-year SDF flood, of which 38 ha lies more than 1 m deep (42 to 46 ha across the plans).

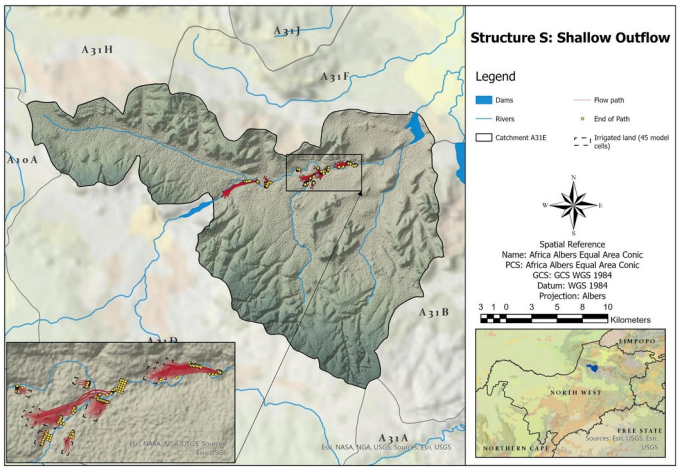


Map 9: Maximum depth of the 1:100-year flood (SDF method) in the modelled reach of the Klein Marico River, from the HEC-RAS 2D model, with the outline of the 1:50-year flood. Hatching shows land mapped as irrigated; the inset shows the irrigated land in more detail. Depth classes 0.01–0.5, 0.5–1, 1–2, 2–4 and >4 m.

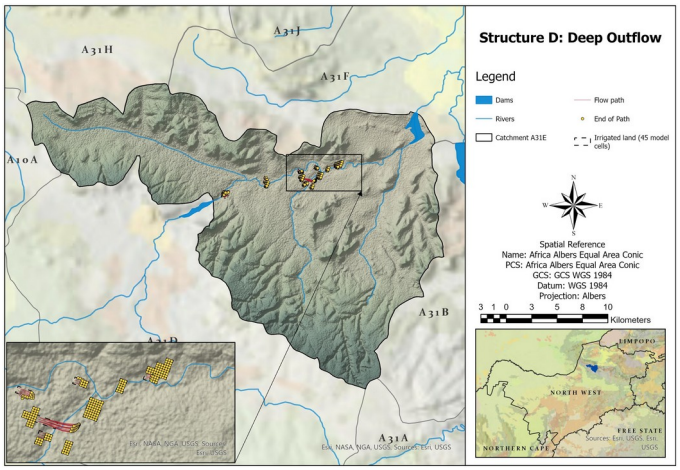
6.8 Pathways and the two waters

In structure S, 387 of the 405 particles (96%) end in a cell where a river drain discharges (Map 10). Their paths have a median length of 537 m, and none is longer than 3.9 km. At an effective porosity of 0.10, the median time to reach the river is 140 years and the 90th percentile about 700 years, while paths from cells beside the river take a few months. In structure D, no particle reaches a river or stream, and 288 of the 405 particles do not leave the cell in which they started, because the water table under the fields stands 15.3 m deep, below the river beds.

(a)



(b)



Map 10: Flow paths of 405 particles released in the 45 model cells under the irrigated land (MODPATH 7; Pollock, 2016), for (a) structure S and (b) structure D. Yellow dots mark the end of each path. In S, 387 of the 405 paths end at a river; in D none do. The paths show advective flow direction only; contaminant transport is not modelled, and travel times depend on an assumed porosity.

The records of the Department show that the two paths carry different water (Figure 15; Table 12). Below Klein Maricopoort Dam, the median EC was 51.2 mS/m in 1989–1993 and 62.3 mS/m in 2012–2023, and 86% to 99% of the samples exceeded the upper limit of the target range for irrigation, 40 mS/m. The three sites along the reach had higher medians of 76 to 83 mS/m in 2004–2007, based on one to three samples each from a different period. The Wilgeboomspruit, which carries water generated inside A31E, had a median of 3.85 mS/m and no exceedance, and the groundwater of A31E has a median of 17.7 mS/m, based on seven samples (DWS, 2021:163). Kromellenboog Dam, which receives both waters, lies between them at 21.7 mS/m. Sulphate follows the same order, with medians of 30 to 57 mg/L below the upper dam and 102 to 145 mg/L along the reach, against 17.7 mg/L in Kromellenboog Dam and

3.05 mg/L in the Wilgeboomspruit. No sample at any site exceeds the guidelines for chloride, sodium or SAR.

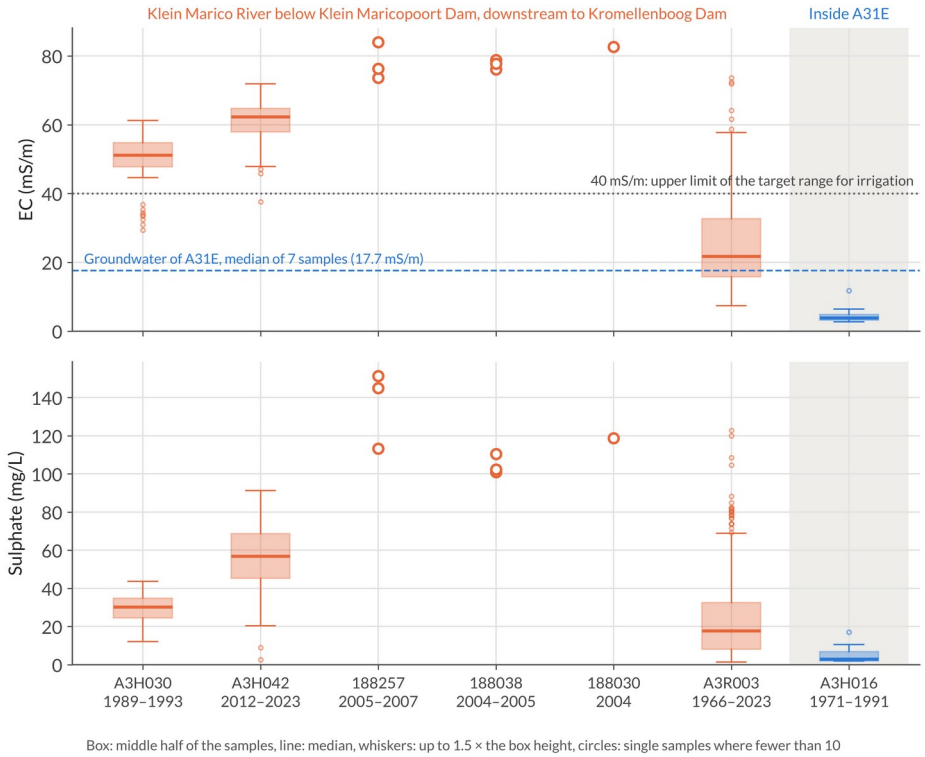


Figure 15: Electrical conductivity (upper panel) and sulphate (lower panel) at the DWS water-quality sites along the Klein Marico River below Klein Maricopoort Dam, in Kromellenboog Dam and in the Wilgeboomspruit (DWS, s.a.e), with the upper limit of the target range for irrigation (40 mS/m; DWAF, 1996a:155) and the median EC of the groundwater of A31E (DWS, 2021). Single samples are shown as circles.

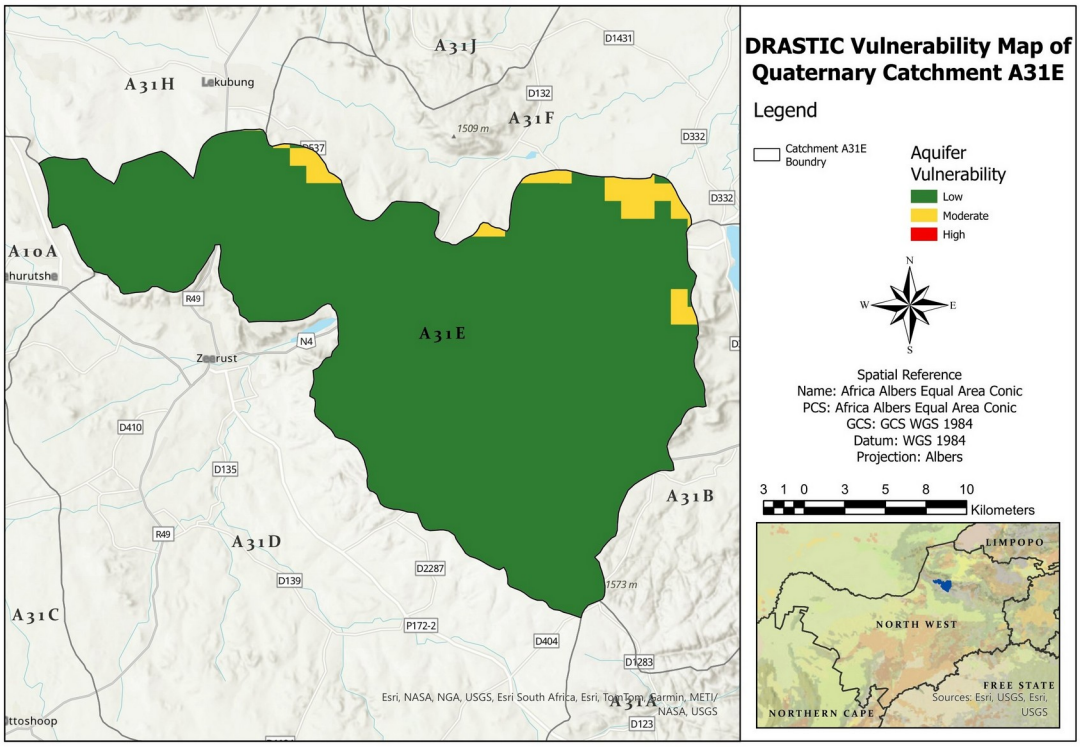
Table 12: Screening of the DWS water-quality records along the two paths (DWS, s.a.e; surface samples). Share = percentage of the samples analysed for that quantity that exceed the guideline value. Guidelines: EC above 40 mS/m, the upper limit of the target range for irrigation (DWAF, 1996a:155); inorganic nitrogen above 0.5 and 2.5 mg/L and phosphate as phosphorus above 0.025 mg/L, the boundaries of the trophic classes, which the guideline defines for the average summer concentration (DWAF, 1996b:84, 97). Chloride, sodium and SAR exceed their irrigation guidelines of 100 mg/L, 70 mg/L and 2.0 (DWAF, 1996a:48, 134, 144) in no sample at any site, and sulphate has no guideline in this screening.

Site	Path	Period	Samples	EC median (mS/m)	EC above guideline	Inorganic N above 0.5 / 2.5 mg/L	Phosphate-P above 0.025 mg/L
A3H030	1	1989–1993	56	51.2	86%	2% / 0%	4%
A3H042Q01	1	2012–2023	81	62.3	99%	12% / 4%	19%
188257	1	2005–2007	3	76.3	100%	0% / 0%	0%
188038	1	2004–2005	3	77.7	100%	0% / 0%	33%
188030	1	2004	1	82.6	100%	100% / 0%	100%
A3R003Q01	1 and 2	1966–2023	248	21.7	17%	16% / 0%	29%
A3H016Q01	2	1971–1991	54	3.85	0%	5% / 2%	20%

Over the common period of 2012–2023, the river below the upper dam had a median EC of 62.3 mS/m and Kromellenboog Dam 20.7 mS/m. With 19% of the inflow to the

lower dam coming from the upper dam, the mixing check gives an EC of about 11 mS/m for the water of the reach, which lies between that of the Wilgeboomspruit and that of the groundwater. The nutrients do not separate the two paths in the same way. Phosphate exceeds 0.025 mg/L in 19% of the samples below the upper dam, in 29% in Kromellenboog Dam and in 20% in the Wilgeboomspruit.

The DRASTIC index is low over 96.7% of A31E and moderate over 3.3% (Map 11; DWAF, cited by Esterhuyse, 2017).



Map 11: Aquifer vulnerability of A31E according to the DRASTIC index (Aller *et al*., 1987), in the classes of the national map: low (below 90), moderate (90–140) and high (above 140) (DWAF, cited by Esterhuyse, 2017). The index is mapped on a 1 km national grid and is not a measurement.

7. Discussion

7.1 The canal supply in drought

The two bounds represent two ways of running short of water, in which a restricted canal fails more often but less deeply. Hashimoto *et al*. (1982:19) found the same trade-off, where “high system reliability was accompanied by high system vulnerability”. A partial delivery allows an irrigator to plant a smaller area or to apply less water, while a dry canal does not. At this dam, restriction costs volume, because much of the water that is held back evaporates, and the rule delivers a smaller share of the registered volume than the unrestricted canal (Table 7). The reliability depends

more on this rule than on the demand, and the rule is the least certain input, since none could be obtained and the two periods of the record give different rules. The test of the balance supports counts of years and volumes more than durations in single months. The record of the canal alone overstates how often the supply failed, because in five of the shortfall years the dam never fell below 75% of its capacity, and the water balance removes this effect by asking for the registered volume in every year.

The capacity of the dam is too small to carry the scheme through a drought that lasts several years. Because evaporation exceeds the rainfall in every month, irrigated crops require water throughout a dry winter of five months. The national register records entitlements and not the actual abstraction. Half of the scheme positions lie on ground that a canal cannot reach, so that the register most probably records a reference point of each property rather than the position of the field. The registered area, the mapped irrigated land and the volume in the canal are quantities of different kinds and cannot be combined, and the results of this study describe the security of an entitlement rather than the water used by crops.

The historic firm yield shows how little of the registered volume the dam can assure. The yield that the Department gives for planning is about four times larger and is also described as a firm yield (DWAF, 2004:54). It rests on hydrology that had not been updated since 1988 and did not include the droughts that followed (DWAF, 2004:83), while the historic firm yield of this study includes the whole drought of 1983 to 1995. A yield for this dam is meaningful only together with the share of months or years in which it is met.

One drought dominates the counts under both bounds, so its cause determines how they should be interpreted. The cause of the low inflow is not known, and two explanations outside A31E are possible. The first is a natural decline in the groundwater stored in the dolomite aquifer above the dam, whose springs declined to almost nil in 1986 and stayed low until 1996 (Pyburn, 2015:28, 44). The second is a greater use of water upstream, since a well field was commissioned in the Zeerust dolomite in 1974, and the Rietpoort Spring stopped flowing about ten years later during a period of low rainfall (Pyburn, 2015:4). That the water of these springs reaches the dam is inferred from the position of the dolomite. After the Millennium Drought in south-eastern Australia, runoff as a fraction of rainfall had likewise not recovered once the drought ended in 37% of the watersheds studied (Peterson *et al*., 2021:745). The

explanations considered plausible there were declining groundwater storage and the harvesting of runoff by small dams (Fowler *et al*., 2022). These match the two explanations above Klein Maricopoort Dam. The records cannot separate the two, which is common where gauged flows are affected by poorly defined developments upstream (Kapangaziwiri *et al*., 2011:128). Since the rainfall of each year does not explain the drought, its recurrence cannot be estimated, and the reliabilities remain frequencies counted in one record. The low inflow is accepted as part of the recorded sequence without a cause being assigned to it.

7.2 Water generated inside A31E and groundwater as a backup

In 1987–1995 the runoff of A31E sustained the scheme below the catchment during the years in which the scheme inside it had almost no water. Since the canal carried 0.30 Mm³/a in these years, little of the net contribution of the reach can have been returned canal water, and it behaves like runoff. The failure of the canal supply was a failure of the water from outside A31E, while the runoff of the catchment continued but entered the river below the canal, where it could not replace the canal supply. Any use of water inside A31E that reduces the flow of the reach, whether it is taken from the river or from groundwater that would otherwise have reached it, reduces the supply of the lower dam. The modelled naturalised runoff shows no reduction during the drought, which agrees with the finding that runoff during a long drought falls below what the earlier rainfall–runoff relationship predicts (Saft *et al*., 2015). In a climate where evaporation exceeds the rainfall in every month, the soils are seldom thoroughly wet, which agrees with the low yield of the Wilgeboomspruit. The Wilgeboomspruit further shows that the store feeding a small stream can empty within three years of below-average rainfall and recover long before the dam does. It is not shown that this store is groundwater, because low flows in South African rivers may also come from storage above the water table (Kapangaziwiri *et al*., 2011:127).

The four water levels do not determine the conductivities of structure D, and each value is an upper limit, since any lower value would fit the levels at least as well. The better fit of structure D comes from the depth of its outflow and not from its conductivities. The regional recharge and transmissivity are not consistent with each other at these boreholes, unless most of the recharge leaves the ground close to where it entered, as the ground drain allows. The archive levels support structure D in

the valleys but decide nothing on the high ground. The water levels in the valleys contradict structure S, while the streamflow contradicts structure D, and the real catchment lies somewhere between the two. The unaccounted 14.5 Mm³/a of the Reserve balance may be taken up by vegetation, may flow out underground or may reflect an overestimate of the recharge. Structures S and D also show what Troldborg *et al*. (2007) found, namely that calibrated models that differ little in their heads can differ greatly in their predictions.

The fact that the shortfall is a small share of the recharge says little about the availability of groundwater, because pumped water is balanced mostly by capture (Section 3.2). Structure S behaves in this way, while structure D gives the opposite answer, because the source of the pumped water depends on where the water table stands before pumping. In structure S it stands above the river beds, so that the rivers gain groundwater and pumping intercepts it. In structure D it stands below the river beds, and there is therefore almost nothing to intercept. A drain cannot add water, so structure D cannot show a river losing water to the aquifer, and its share of less than 1% is the least the model allows. The depth of the outflow determines the source of the pumped water, while the conductivity determines the drawdown, which the calibration did not fix. In structure S, an abstraction under the fields is largely a transfer of water that would otherwise have flowed to Kromellenboog Dam. The water that boreholes can draw on lies mainly near the surface, where it is the first to fail during a drought (Calow *et al*., 2010). In the terms of Calow *et al*. (2010), groundwater in A31E is limited as a backup not by its availability, but by access to it and by the question of whose water is taken.

7.3 Climate change

The difference between the two climate cases is set by the rainfall rather than by the warming. Both cases include the same warming of 1.9 °C, which raises the evaporative demand by a few per cent (Scheff & Frierson, 2014). The change in rainfall from −3.1% to −8.5% becomes a change in inflow from −7% to −20%. This amplification agrees with the finding that a change in rainfall is magnified in river flow where about 500 mm of rain falls per year (De Wit & Stankiewicz, 2006). It also explains why the rainfall, which is uncertain even in sign, determines the outcome more than the warming does. A wetter outcome offers little relief, because the wettest

projections give little more rain than the record, and in wet years most of the additional inflow spills from the dam.

The uniform factors also leave out what the projections emphasise. These include longer and more frequent droughts (Trisos *et al*., 2022; Ranasinghe *et al*., 2021:1771), a later start of the rains, which would lengthen the months in which the canal draws on the dam before the first inflow arrives, and runoff that falls disproportionately during long droughts (Saft *et al*., 2015). A uniform factor keeps the recorded sequence of years, and the inflow factor leaves out the effect of warming on the catchments above the dam. Cases M and D are more likely to understate the loss of reliability than to overstate it. Groundwater recharge in drylands depends mainly on episodes of heavy rainfall (Cuthbert *et al*., 2019), so that a drier climate with fewer such episodes could reduce it by more than the factor used. Under the pumped cells of structure D, the abstraction has drawn the water table below the ground drain, so that the drain no longer holds it, and the drier climate adds directly to the drawdown. The climate cases increase the shortfall while the water on which an abstraction would draw decreases, and the groundwater backup weakens as the need for it grows.

7.4 Flood risk in this reach

The design floods of the reach are uncertain mainly because of the choice of method. Nearly half of the catchment above the reach is underlain by dolomite, all of it above Klein Maricopoort Dam, and the SDF is probably an overestimate here (SANRAL, 2013:3-45). The Midgley and Pitman peak is consistent in direction with the warning of the manual that the flood of a dolomitic catchment may be less than half of the SDF. No method can be preferred on the evidence, because a statistical analysis requires an adequate record of flood peaks (Section 3.4), and the available series are too short or regulated. The river gauge at the dam wall has 15 yearly peaks since 2012 (largest 25 m^3/s), and the gauge below Kromellenboog Dam has 47 since 1980 (largest at least 81 m^3/s) (DWS, s.a.b). Both are regulated by dams and mostly record years with almost no flow. They cannot choose between the methods, and no flood near the design floods of Table 11 has been recorded at the lower gauge. The flooded land is better defined than the flood itself, because the valley confines the flow and a larger flood is carried mainly by a greater depth. The irrigated land lies on this floodplain, so that its users face two risks of a different kind, namely a shortage of water that lasts for

years and the flooding of their land. Such a flood is also a pathway along which soil, and whatever lies on it, can reach the river.

Because the reach passes a flood on almost unchanged, a flood that leaves the upper dam continues with nearly its full peak towards Kromellenboog Dam, together with the floods of the tributaries, which the model does not add. The Department rated the spillway of the upper dam as “a major risk” and the spillway capacity of the lower dam as “a major risk factor” (DWAF, 2004:170–171), and these ratings predate the rehabilitation of the upper spillway from 2009 (News24, 2011). The flooded extent of this study is a screening and not a flood line. The model does produce 1:50-year and 1:100-year outlines, but the National Water Act requires that the layout plan of a new township shows “lines indicating the maximum level likely to be reached by floodwaters on average once in every 100 years” (section 144, quoted by SANRAL, 2013:11-4). Such lines decide where development may take place, and they would require terrain accurate to well below a metre, a surveyed channel, a design flood that can be preferred on evidence and a model checked against a recorded flood, none of which were available to this study.

7.5 What the two waters mean for the canal users, the aquifer and the lower dam

The water released from Klein Maricopoort Dam lies above the target range for irrigation in most samples, and the canal, which is fed from the same dam, probably carries water of the same quality. This water exceeds the irrigation target in its general salt content but not in sodium in particular. It is more saline because it comes from outside A31E and probably carries the return flows of Zeerust and of the irrigation and mines above the dam (DWAF, 2004:57). Evaporation also removes a third of the water that enters the dam but leaves its salts behind. Whether this salinity limits the crops grown depends on the crops and on how the water is applied, which the records do not show.

Canal water with a median EC of about 62 mS/m is applied on land above groundwater whose median EC in A31E is 17.7 mS/m in seven samples (DWS, 2021:163). The water that drains below the fields carries salt into the aquifer. In structure S, this water reaches the rivers decades to centuries later, and an abstraction under the fields would capture the same water that carries whatever drains from the fields to the river. In

structure D, the water stays close to the fields and leaves by evaporation and uptake by plants, which would concentrate its salts further. In either structure, the aquifer that this study assesses as a backup to the canal is the same aquifer that receives the drainage of the fields, and its quality is not monitored. Kromellenboog Dam, whose water is kept relatively fresh by the runoff of A31E, would also lose part of this fresh water in structure S if groundwater were abstracted under the fields, but the model does not represent this effect and it remains an inference.

The mixing check combines the share of 1987–2009 with salinities of 2012–2023 and ignores the concentration of salts by evaporation in the lower dam, so that the records are consistent with mixing but do not prove it. Episodes of nutrient-rich water occur on both paths, including in the runoff of A31E itself.

7.6 The answer to the main research question and its limits

The main research question consists of three parts. Firstly, the canal supply is not reliable against the registered volume, and one drought of more than a decade dominates the shortfall. Secondly, groundwater inside A31E could replace the average shortfall in volume, but the study cannot determine whether this water would be new water or would be taken from the rivers that feed Kromellenboog Dam. During a drought like the long one, the required abstraction could not be delivered from under the irrigated land in one structure, and it would come mostly from the rivers in the other. Thirdly, under the conditions projected for about 2050, the reliability decreases further, while the groundwater answer keeps its form but weakens in degree.

Each bound has a known cause and could be narrowed by specific information. The difference between the two bounds of the canal supply is caused by the unknown operating rule, and the difference between the two model structures by the unknown position of the water table relative to the river bed. The range of the drawdown depends on the conductivity and thickness of the valley deposits. The interaction between groundwater and surface water rests on an inferred indication, two model structures and a mixing check rather than on a demonstration, because no series of water levels or low flows exists along the river.

The data did not allow more. The inflow to the dam is not gauged and cannot be tested, and the canal is not measured beyond its head. No water-level series exists, so that the groundwater model has no time, and no pumping tests exist, so that it cannot show

how many boreholes would be needed or what they would yield. The thickness of the surface deposits is unknown, the single layer applies their conductivity over its full thickness of 207.5 m, and the western arm, a quarter of A31E, has no static water level. No flood peak has been recorded in the reach, and the daily record of the dam shows no spill during the flood warning of January 2011 (DWS, s.a.b; News24, 2011), so that neither the flood methods nor Manning's n could be tested against a flood. The 30 m terrain of 2000 has heights that may be several metres in error (Farr *et al*., 2007). The effective porosity is not known, the water-quality records cover different periods, three sites have one to three samples, and no series of groundwater quality exists.

8. Recommendations

Each recommendation names its addressees and goes no further than the result on which it is based.

1. Plan for shortfalls that last several years (DWS; Premier's Office). The present dam delivers the registered volume in full in 45% to 77% of the years, and in 32% to 70% under the conditions projected for about 2050, while the wettest projections offer no counterweight. Drought plans should treat the registered volume as a volume that the dam cannot assure, and they should be tested against the dry end of the projections rather than the median (Sections 6.3, 6.6 and 7.3).
2. Decide on the restriction rule deliberately and make it known to the irrigators (DWS). The rule affects the reliability more than the demand does, and the water balance of this study can be used to compare candidate rules before one is adopted (Section 7.1).
3. Regard groundwater as a partial backup at most, and assess any abstraction in the valley for its effect on the river (DWS; Premier's Office; AGES). The catchment-scale balance of recharge and use is no evidence that groundwater is available to the scheme, since in one structure of the model about nine-tenths of an abstraction is taken from the rivers that supplied 98% of the inflow to Kromellenboog Dam during the drought. If boreholes are investigated, they should be spread over the valley deposits rather than concentrated under the fields, which lowers the drawdown many times in both structures (Sections 6.5 and 7.2).

4. Determine the 1:50-year and 1:100-year flood lines for the Klein Marico River between the two dams (DWS; Premier's Office). These lines and the present state of both spillways should be known before further development takes place on the floodplain (Section 7.4).
5. Monitor the quality of the water at the head of the canal and in boreholes below the irrigated land (DWS; AGES). The canal has never been sampled, and no record of groundwater quality exists under the fields, whose drainage enters the aquifer that was assessed as a backup (Section 7.5).
6. Collect the data that would narrow the answer most (all addressees). The first four priorities, in order, are:
   (a) The operating and restriction rule of the scheme, with yearly records of the water ordered and delivered.
   (b) Water levels and electrical conductivity in boreholes beside the Klein Marico River below the dam.
   (c) The archive's lithological logs for the valley deposits, and pumping tests in those deposits.
   (d) Water-level series in the valley and in the western arm.

9. Conclusions

The records show that the canal supply from Klein Maricopoort Dam has failed for long periods. After the dam was raised, the canal carried less than the registered volume in almost two-thirds of the countable years, and for ten successive years from 1984 it carried little water. The inflow to the dam stayed low for thirteen years although the rainfall had recovered, for a reason that probably lies upstream but has not been established. The water generated within A31E decreased far less during the drought and sustained the scheme downstream, but it enters the river below the canal and could not replace the canal supply.

A monthly water balance of the dam, tested against the records of 1987–2009, shows that the present dam delivers the registered volume in full in just under half to three-quarters of the years. Without restriction, it could have supplied about 15% of that volume in every month of the record. The result depends more on whether deliveries are restricted as the dam falls than on the demand, and this rule is not known.

A groundwater flow model in two structures shows that an abstraction equal to the mean shortfall is small compared with the recharge and can be delivered in both structures. The structures disagree on the drawdown and on whether the water comes from the rivers or from a deep outflow, and four static water levels cannot decide between them.

Under the warmer and probably drier conditions projected for about 2050, the reliability decreases and the shortfalls become longer, modestly at the median of the projections and markedly at their dry end, and these cases probably understate the loss. The discharge of groundwater to the rivers decreases by about as much as the recharge.

The 1:100-year flood passes through the modelled reach almost unchanged and covers about 40% of the cultivated land in it, whichever method is used. The water released from the dam exceeds the salinity target for irrigation in most samples, unlike the fresh runoff of A31E, and the aquifer assessed as a backup to the canal is the same aquifer that receives the drainage of the fields.

The canal supply to A31E is not reliable against the volume registered for it, and its main weakness is a drought of many years whose recurrence cannot be given a probability. Groundwater could offset the average shortfall in volume, but it cannot yet be determined whether this water would be new water or the river’s own, or whether it could sustain the scheme through such a drought.

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AI Declaration Page

Title of the work: Storage, drought and climate change: the reliability of the canal supply from Klein Maricopoort Dam and the role of groundwater in A31E, North West Province.

Author: LH Stroebel (student number 42766613). Supervisor: Sascha Roopa.

1. Was AI used? Yes.

2. Purpose. AI was used to process data and build the dam water-balance spreadsheet from the author's records, to draw the graphs, to draft and edit text, to structure the document and to check facts and references.

3. Tool used. Claude (Anthropic).

4. How AI was applied in the research process. The author supplied the records, defined the study and made the decisions on method. Claude processed the records and built the dam water-balance spreadsheet from them, drew the graphs, drafted and edited text and structured the document. The Introduction and the first version of the Study Area were written by the author, and Claude then edited the text and extended it to cover the whole project. The author ran the GMS/MODFLOW, MODPATH and HEC-RAS models, made the maps in ArcGIS Pro and checked the outputs against the records.

5. How the accuracy and reliability of the AI outputs were verified. The author checked the model outputs against the records. Claude was also used to check facts and references against the sources. The full log of the work with Claude is available on request.

Signature: _____

Date: _____